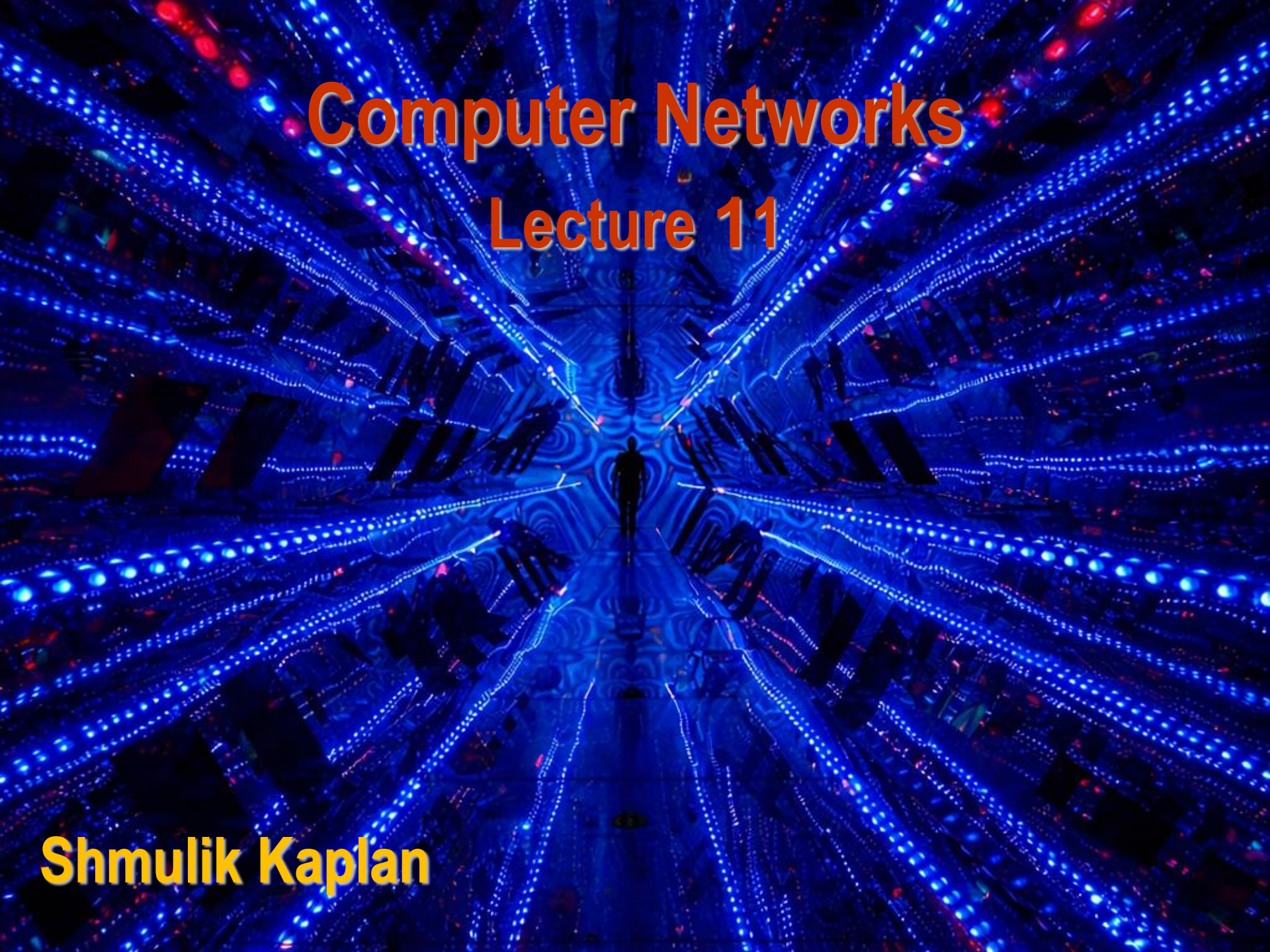
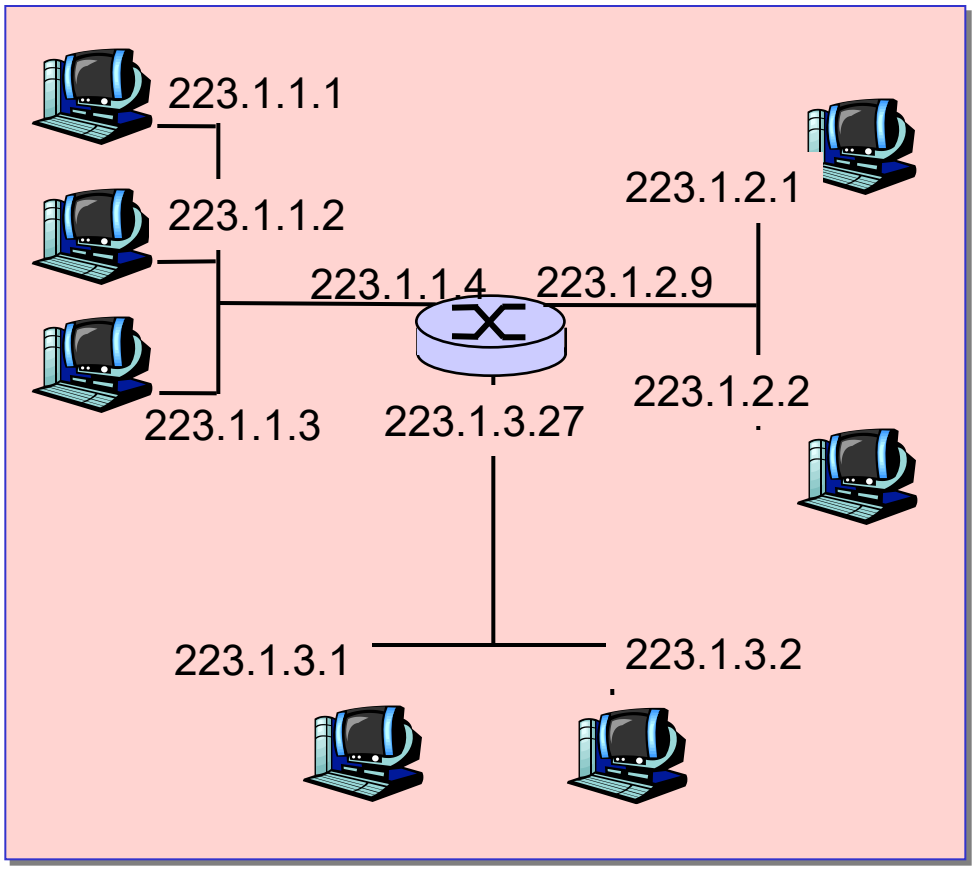

Computer Networks
Lecture 11
Shmulik Kaplan

Today

- IP addressing introduction
- IP addressing CIDR
- How routing works over IP
- IP in details
- ICMP
  - Ping
  - Traceroute
- DHCP
- NAT
- IP v6

IP Addressing: introduction

- IP address: 32-bit identifier for host, router *interface*
- *interface:* connection between host, router and physical link
  - router's typically have multiple interfaces
  - host may have multiple interfaces
  - IP addresses associated with interface, not host, router

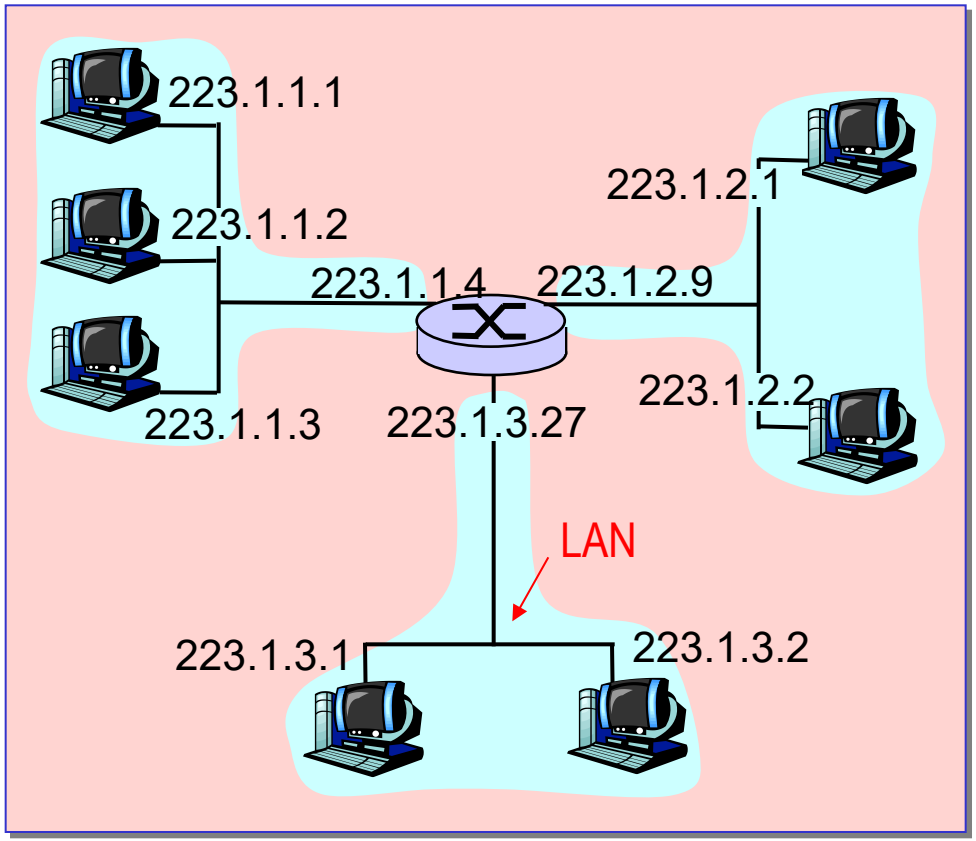


223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

IP Addressing

network part (high order bits) + host part (low order bits)

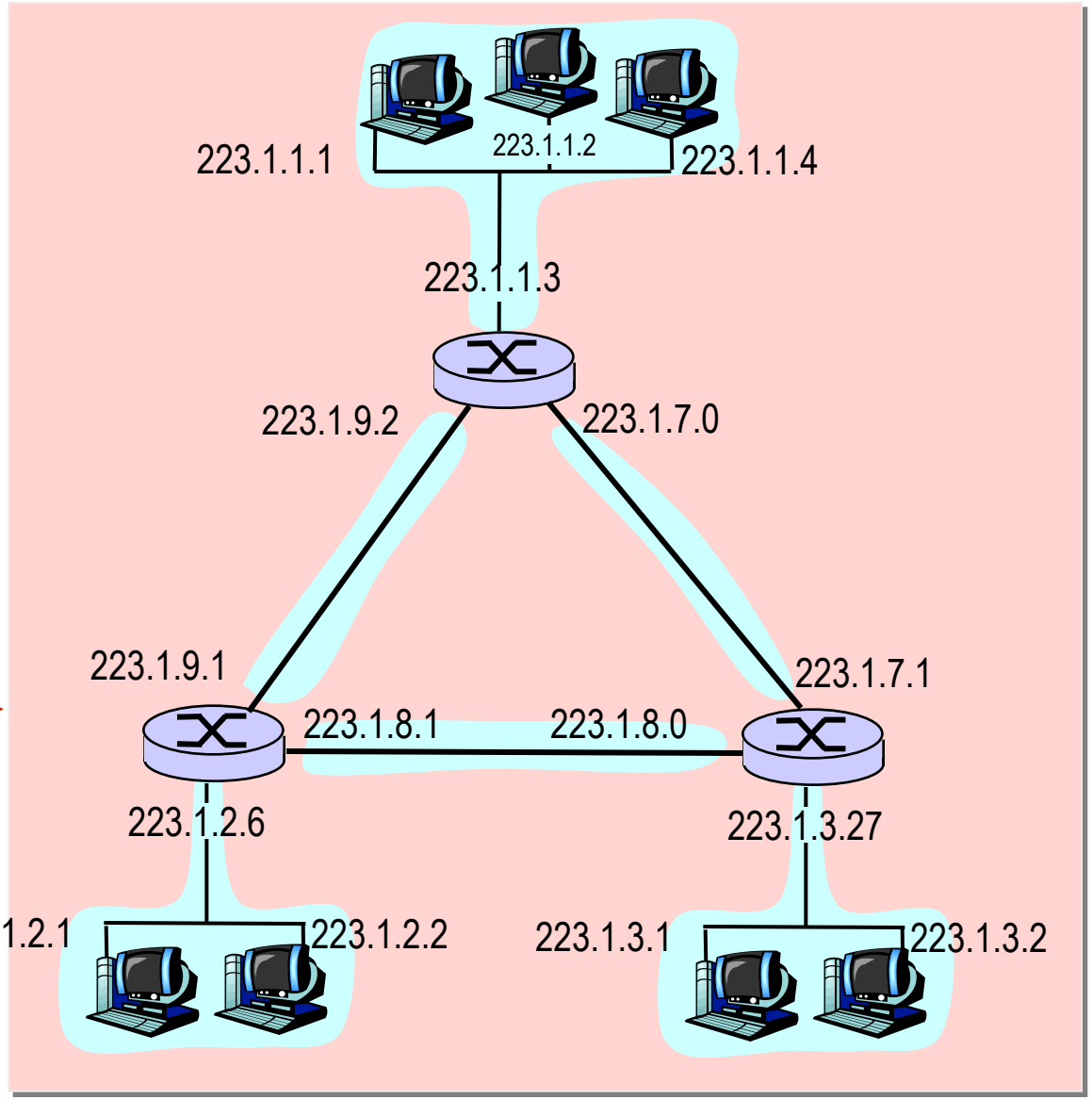


IP Addressing

How to find the networks?

- Detach each interface from router, host
- create “islands of isolated networks”

Interconnected system consisting of six networks



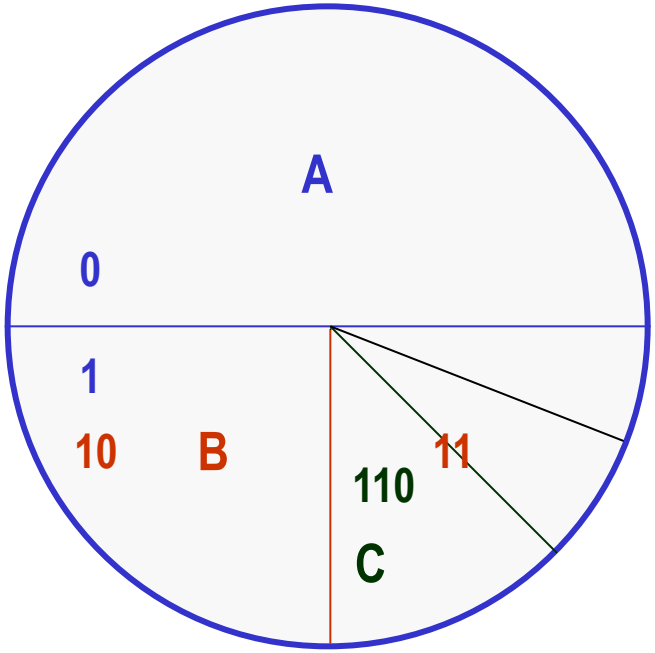
IP Addresses

"class-full" addressing

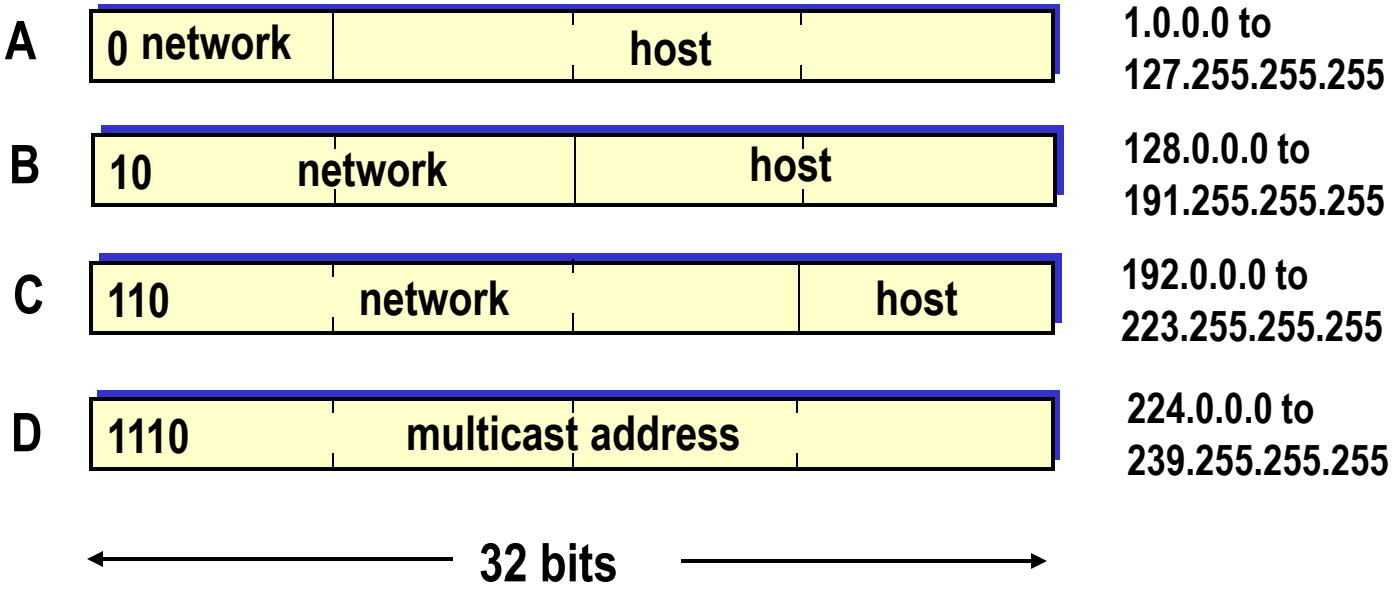
A B C

of networks:

Size of network:



class



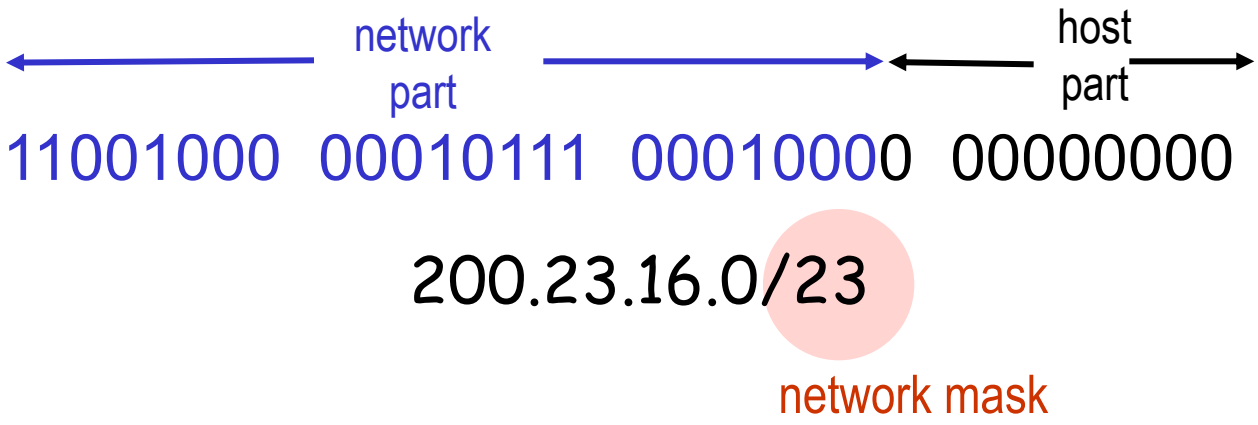
Today



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IP addressing: CIDR

- Classful addressing:
  - inefficient use of address space, address space exhaustion
  - e.g., class B net allocated enough addresses for 65K hosts, even if only 2K hosts in that network
- CIDR: Classless InterDomain Routing
  - network portion of address of arbitrary length
  - address format: a.b.c.d/x, where x is # bits in network portion of address



IP addresses: how to get one?

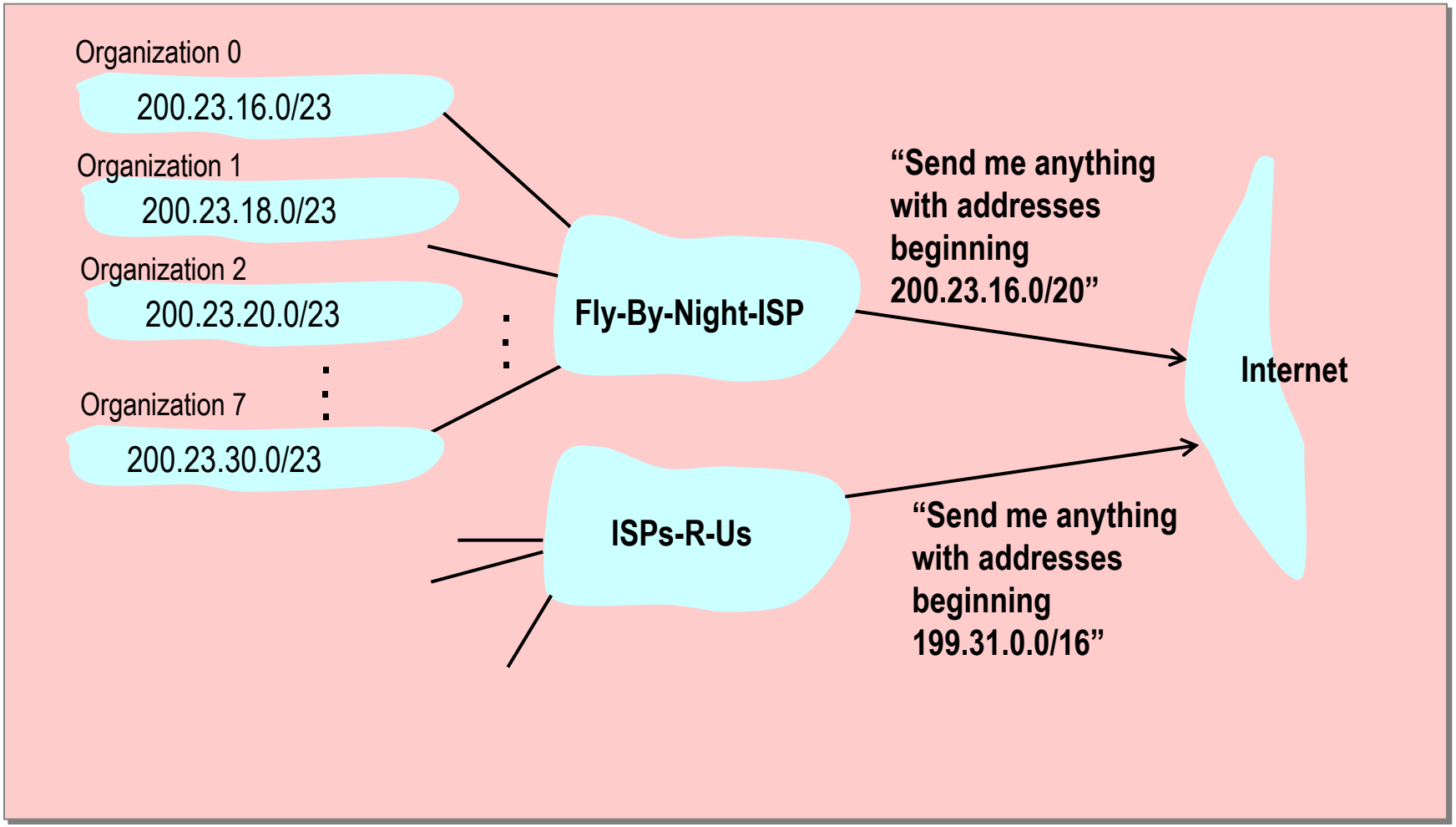
Network (network portion):

- get allocated portion of ISP's address space:

ISP's block	**11001000 00010111 00010000**	**00000000**	**200.23.16.0/20**
Organization 0	**11001000 00010111 00010000**	**00000000**	**200.23.16.0/23**
Organization 1	**11001000 00010111 00010010**	**00000000**	**200.23.18.0/23**
Organization 2	**11001000 00010111 00010100**	**00000000**	**200.23.20.0/23**
…	**…..**	**….**	**….**
Organization 7	**11001000 00010111 00011110**	**00000000**	**200.23.30.0/23**

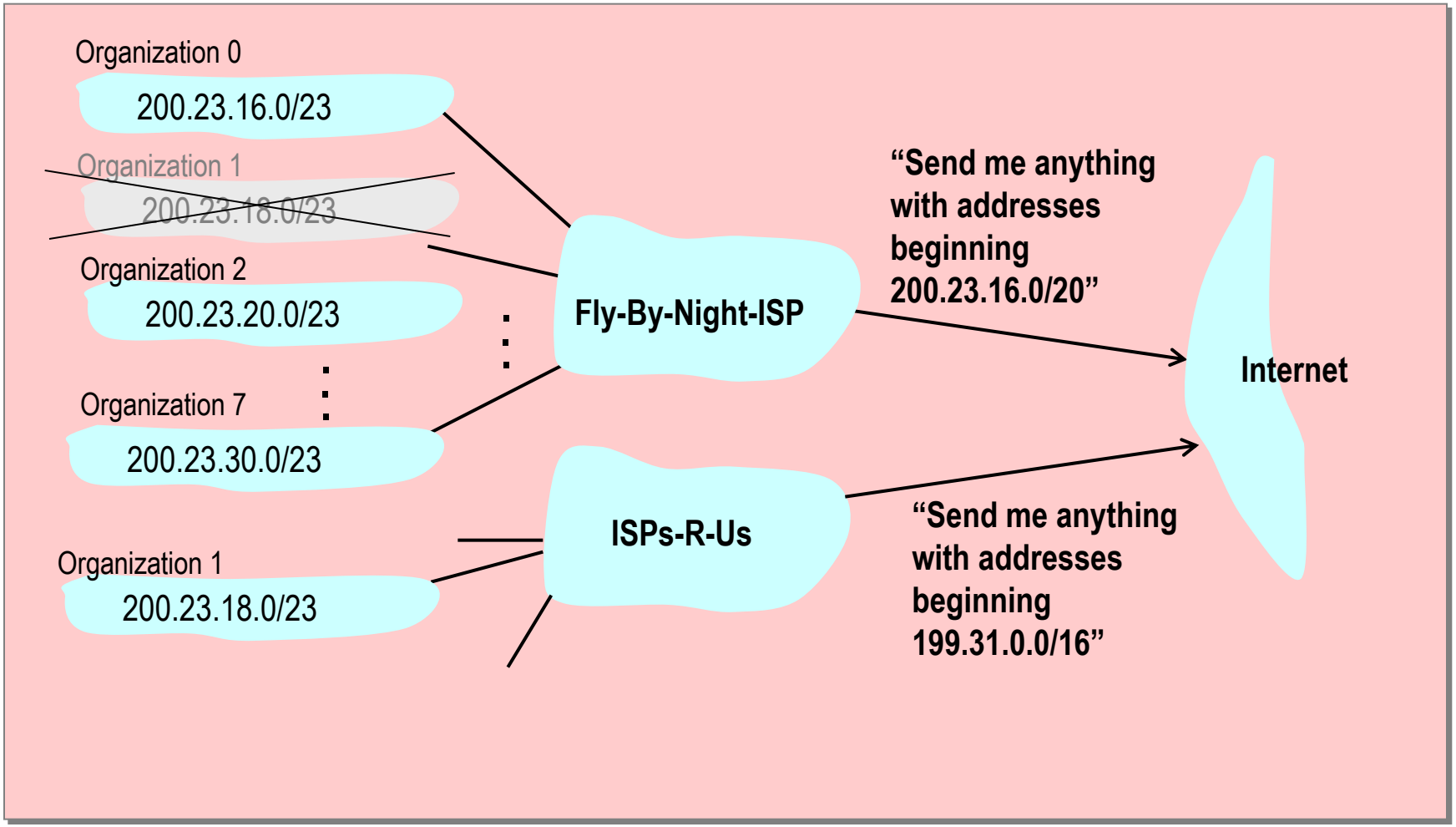
Hierarchical addressing: route aggregation

Hierarchical addressing allows efficient advertisement of routing information



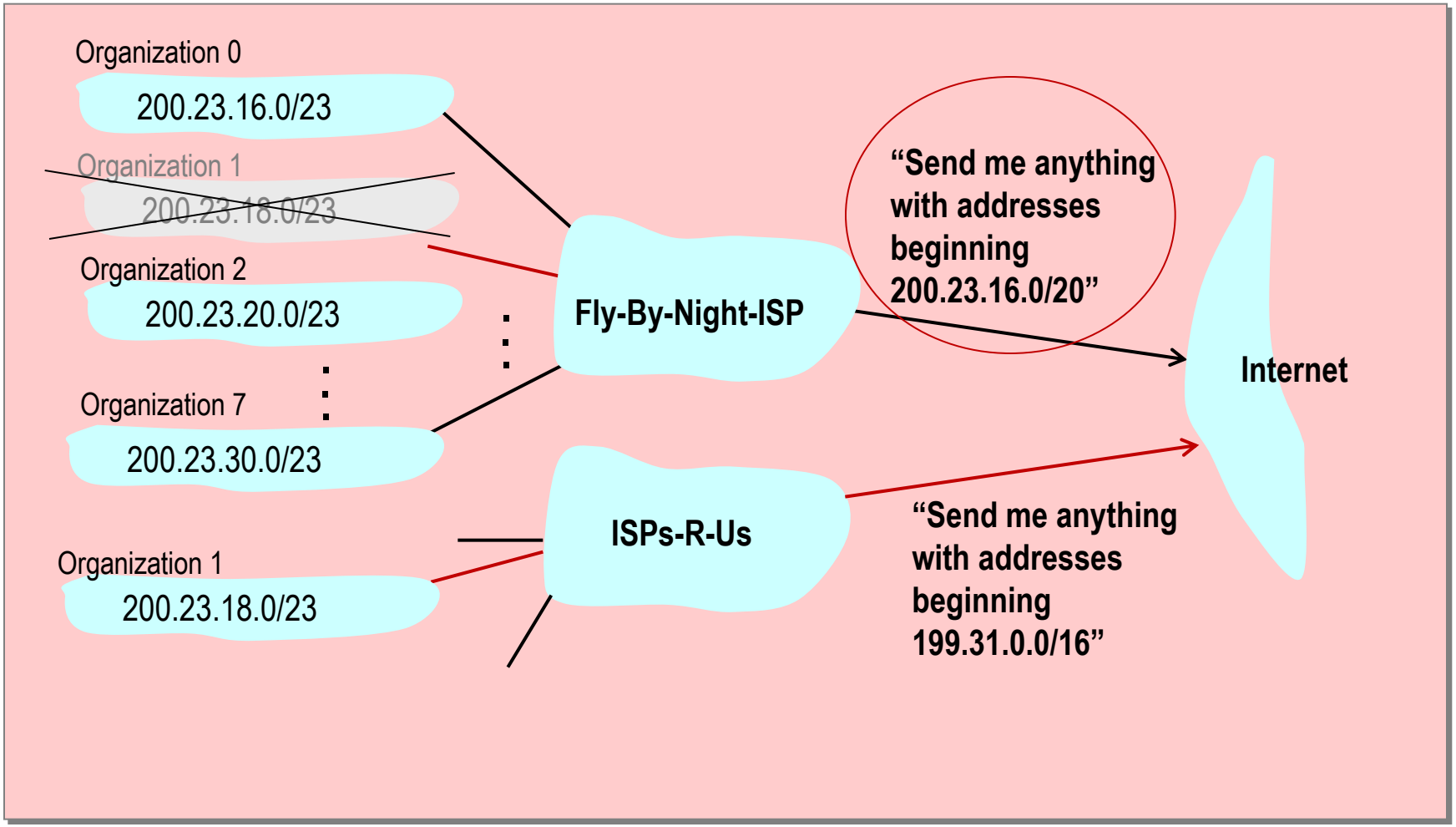
Hierarchical addressing: route aggregation

Hierarchical addressing allows efficient advertisement of routing information



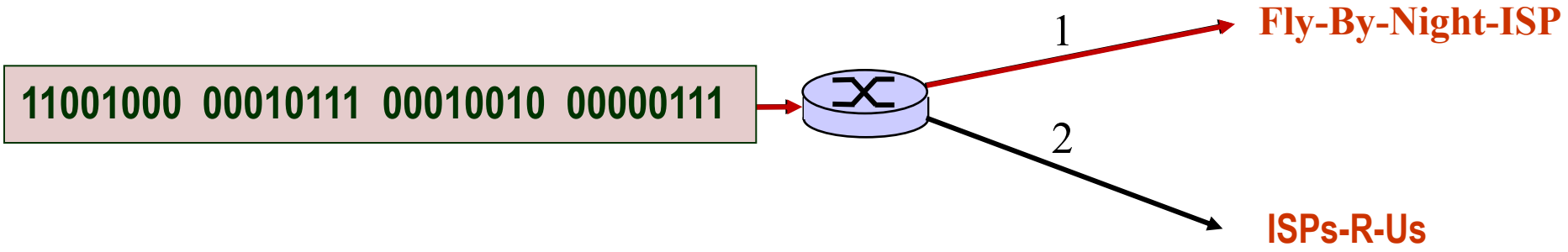
Hierarchical addressing: route aggregation

Hierarchical addressing allows efficient advertisement of routing information



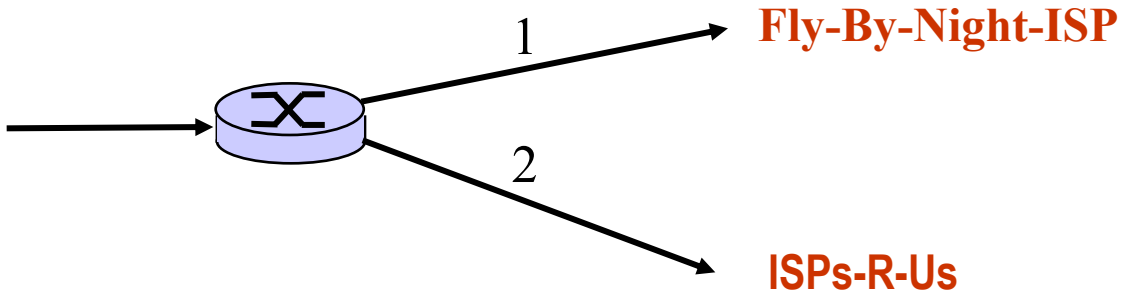
Longest prefix – before migration

Fly-By-Night-ISP	11001000 00010111 00010000 00000000 200.23.16.0/20	1
ISPs-R-Us	11000111 00011111 00000000 00000000 199.31.0.0/16	2



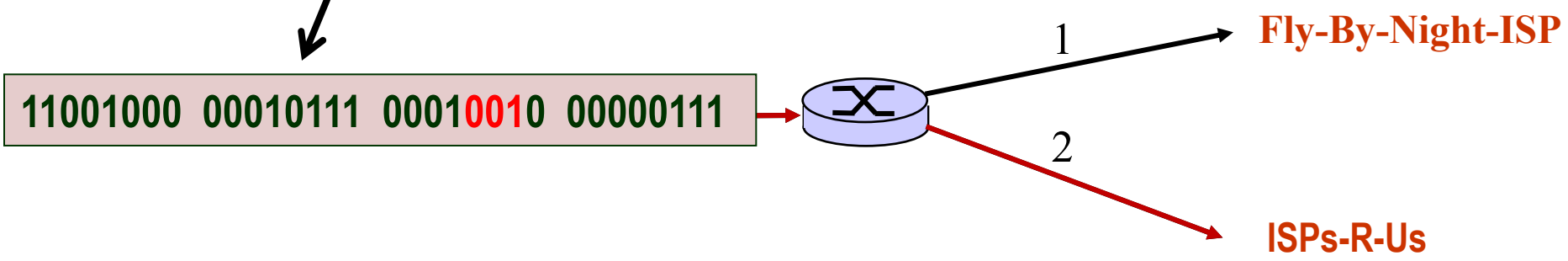
Longest prefix - changes

Fly-By-Night-ISP	11001000 00010111 00010000 00000000 200.23.16.0/20	1
ISPs-R-Us	11000111 00011111 00000000 00000000 199.31.0.0/16	2



Longest prefix - works

Fly-By-Night-ISP	11001000 00010111 00010000 00000000 200.23.16.0/20	1
ISPs-R-Us (Organization 1)	11001000 00010111 00010010 00000000 200.23.18.0/23	2
ISPs-R-Us	11000111 00011111 00000000 00000000 199.31.0.0/16	2



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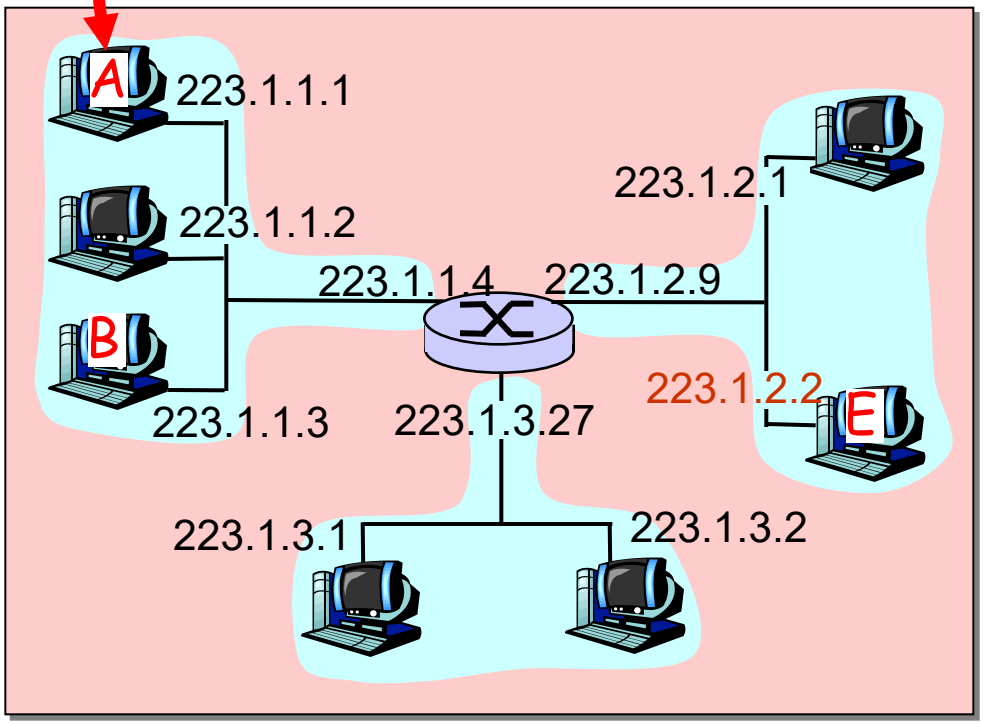
Getting a datagram from source to dest.

misc fields	223.1.1.1	223.1.2.2	data

Starting at A, dest. E

- look up network address of E
- E on *different* network
  - A, E not directly attached
- routing table: next hop router to E is 223.1.1.4
- link layer sends datagram to router 223.1.1.4 inside link-layer frame
- datagram arrives at 223.1.1.4
- continued…..

Dest. Net.	next router	Nhops
223.1.1		1
223.1.2	223.1.1.4	2
223.1.3	223.1.1.4	2



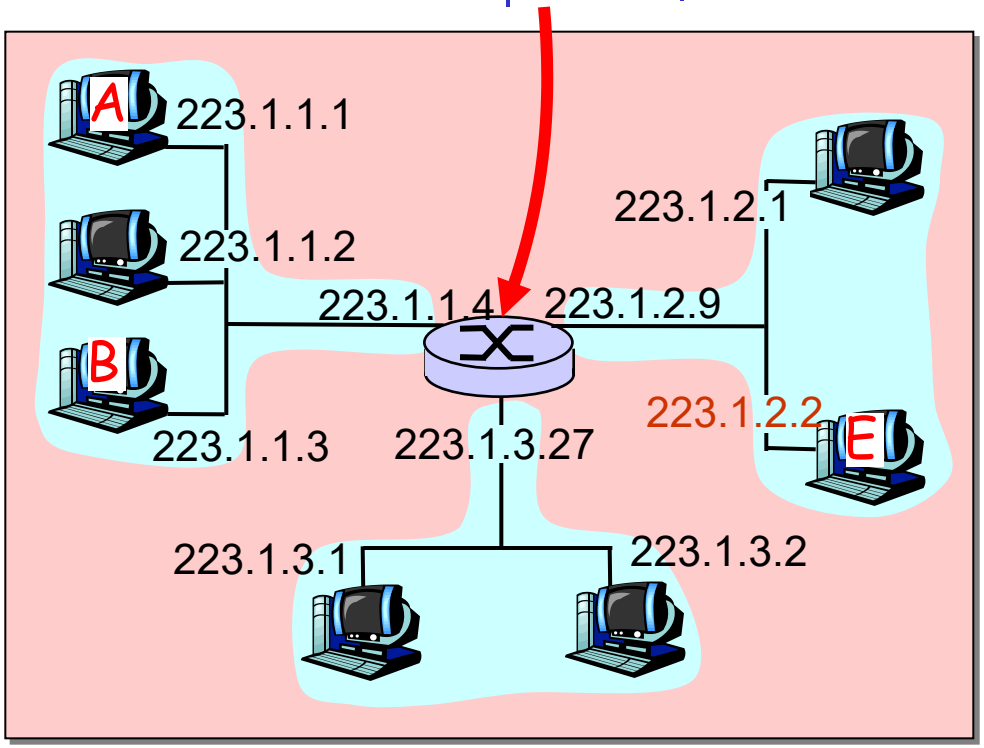
Getting a datagram from source to dest.

misc fields	223.1.1.1	223.1.2.2	data

Dest. network	next router	Nhops	interface
223.1.1	-	1	223.1.1.4
223.1.2	-	1	223.1.2.9
223.1.3	-	1	223.1.3.27

Arriving at 223.1.4, destined for 223.1.2.2

- look up network address of E
- E on *same* network as router's interface 223.1.2.9
  - router, E directly attached
- link layer sends datagram to 223.1.2.2 inside link-layer frame via interface 223.1.2.9
- datagram arrives at 223.1.2.2!!!



Example: IP addressing and routing

- ~ 1Million of addresses are available starting at 194.24.0.0/12
- Cambridge U. asks for 2048 addresses, then
- Oxford U. asks for 4096 addresses, then
- Edinburgh U. asks for 1024 addresses.

- **How to assign the addresses? You have to assign addresses immediately as the request arrives.**

Example: IP addressing and routing

Starting at 194.24.0.0; Cambridge 2048; Oxford 4096; Edinburgh 1024.

University	First address	Last address	How many	Mask
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12.0/22
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20

Address	Mask
C: 11000010 00011000 00000000 00000000	11111111 11111111 11111000 00000000
C: 11000010 00011000 00000111 11111111	
E: 11000010 00011000 00001000 00000000	11111111 11111111 11111100 00000000
E: 11000010 00011000 00001011 11111111	
O: 11000010 00011000 00010000 00000000	11111111 11111111 11110000 00000000
O: 11000010 00011000 00011111 11111111	

Example: IP addressing and routing

Address	Mask	Link
C: 11000010 00011000 00000000 00000000	11111111 11111111 11111000 00000000	3
E: 11000010 00011000 00001000 00000000	11111111 11111111 11111100 00000000	3
O: 11000010 00011000 00010000 00000000	11111111 11111111 11110000 00000000	0

A packet comes in addressed to 194.24.17.4 , in binary:

11000010 00011000 00010001 00000100

First, it is ANDed with Cambridge mask to get

11000010 00011000 00010000 00000000 does not match

Next, it is ANDed with Edinburgh mask to get

11000010 00011000 00010000 00000000 does not match

Next, it is ANDed with Oxford mask to get

11000010 00011000 00010000 00000000 match

Aggregate entry 194.24.0.0/20

11000010 00011000 00000000 00000000 11111111 11111111 11110000 00000000 3

Longest prefix matching

Prefix Match	Link Interface
11001000 00010111 00010	0
11001000 00010111 00011	1
11001000 00010111 00011000	2
otherwise	3

Examples

DA: 11001000 00010111 00010110 10100001 Which interface?

DA: 11001000 00010111 00011000 10101010 Which interface?

IP addressing: the last word...

Q: How does an ISP get block of addresses?

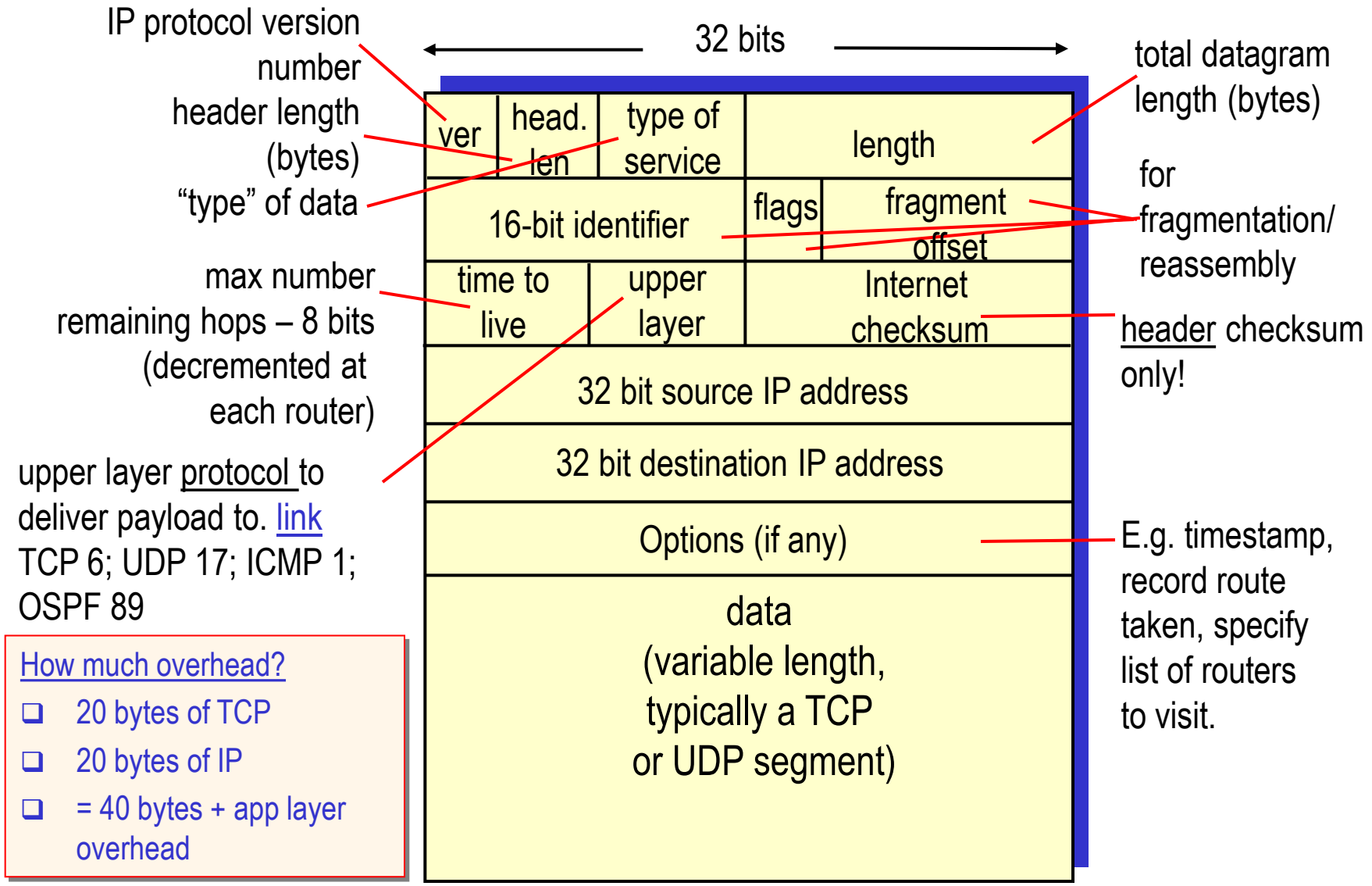
A: ICANN: Internet Corporation for Assigned Names and Numbers

- allocates addresses
- manages DNS
- assigns domain names, resolves disputes

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IP datagram format

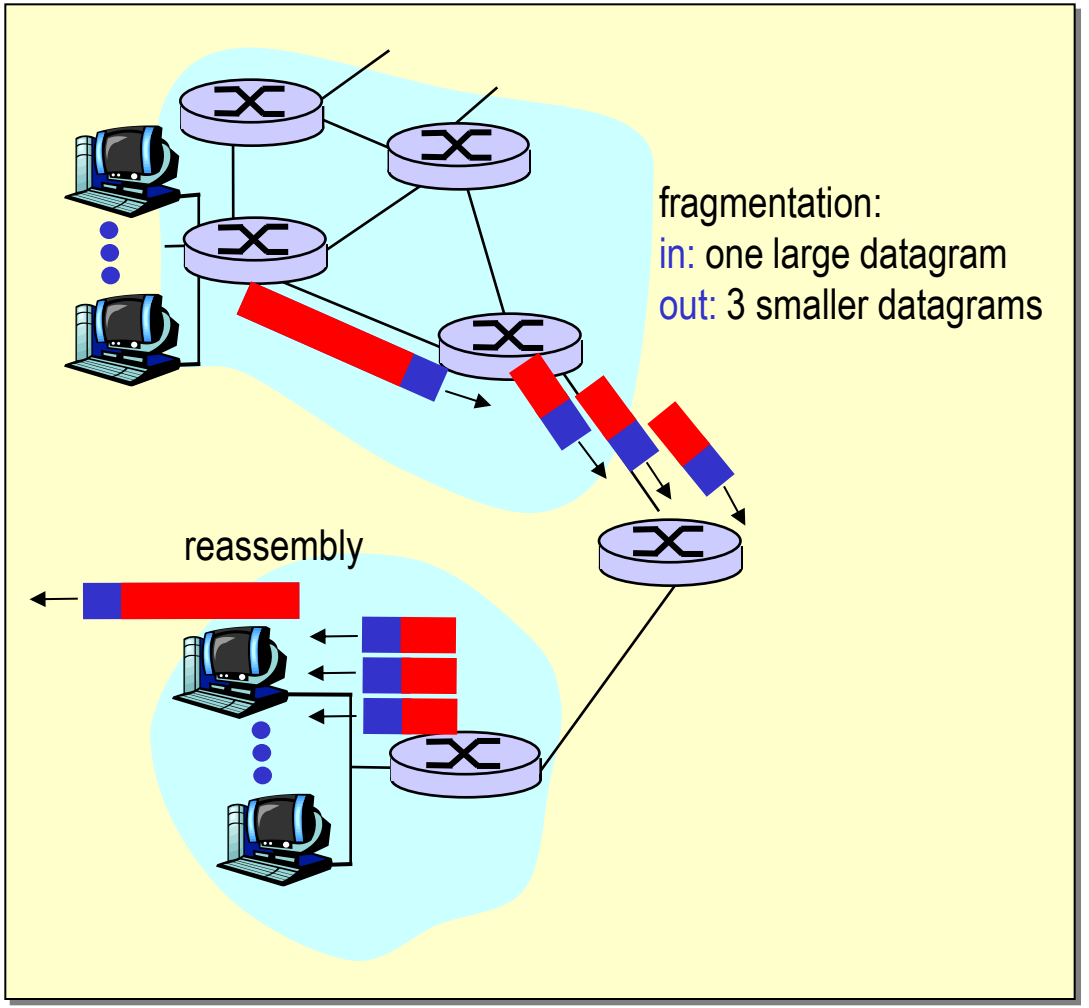


IP header checksum

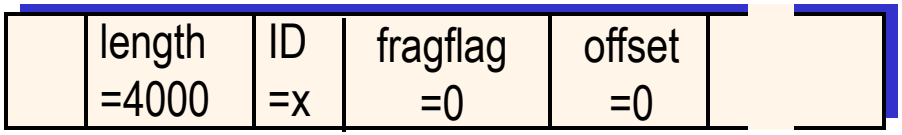
- *Before transmitting the packet, the sender computes a 16-bit checksum of the bits in the IP header and includes the result in the checksum field.*
- *Routers alone the path update the header checksum, to account for any changes in the header, such as decrementing the TTL field.*
- An IP packet with incorrect header checksum is discarded by the receiving machine/router.
- The IP checksum applies only to IP headers and not to the contents of the packets.
- Checking for corrupted data can be provided by the higher-level protocol, such as UDP or TCP, with a separate checksum.

IP Fragmentation & Reassembly

- network links have MTU (max.transfer size) - largest possible link-level frame.
  - different link types, different MTUs
- large IP datagram divided ("fragmented") within net
  - one datagram becomes several datagrams
  - "reassembled" only at final destination
  - IP header bits used to identify, order related fragments



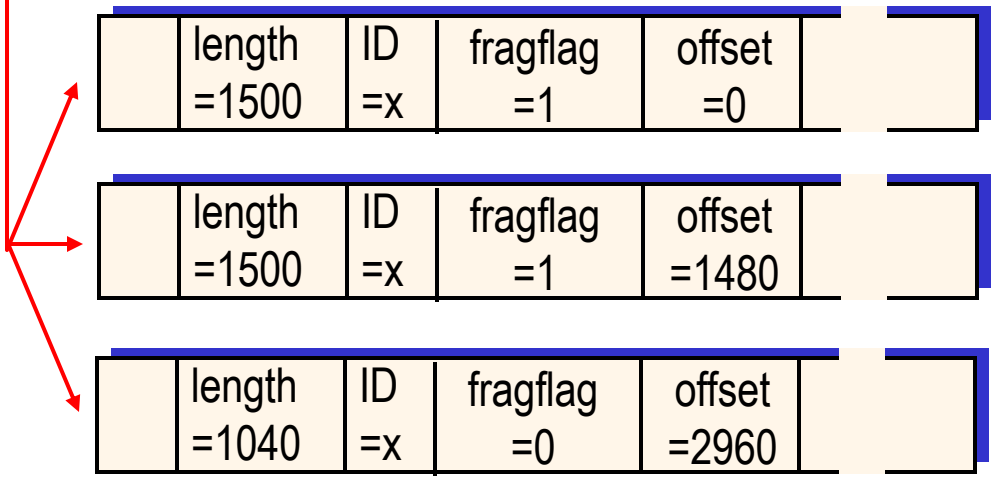
IP Fragmentation and Reassembly



Example

- 4000 byte datagram
- MTU = 1500 bytes

One large datagram becomes several smaller datagrams



Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

כאשר חבילת IPv4-(datagram) מגיעה לנתב (router), הנתב צריך לחשב מחדש את ה- checksum .

1. נכון
2. לא נכון

When IPv4 datagram reaches a router, the router needs to calculate again the checksum.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

נתבים נותנים בדרך כלל עדיפות לחבילות UDP לעומת חבילות TCP.

1. נכון
2. לא נכון

Routers usually give higher priority to UDP packets over TCP packets.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

נתב יעביר חבילה לכוון היעד על ידי שימוש בכניסה הראשונה בטבלת הניתוב המתאימה לכתובת היעד של החבילה.

1. נכון
2. לא נכון

A router will forward a packet towards a destination using the first routing table entry that matches the packet's destination address.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

לכל נתב ברשת יש בדיוק כתובת IP אחת.

1. נכון
2. לא נכון

Each router in the network has exactly one IP address.

1. True
2. False

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ICMP: Internet Control Message Protocol

- used by hosts, routers, gateways to communication network-level information
  - error reporting: unreachable host, network, port, protocol
  - echo request/reply (used by ping)
  - Traceroute also uses IMCP msgs
- network-layer "above" IP
  - ICMP msgs carried in IP datagrams
- ICMP message: type, code plus first 8 bytes of IP datagram causing error

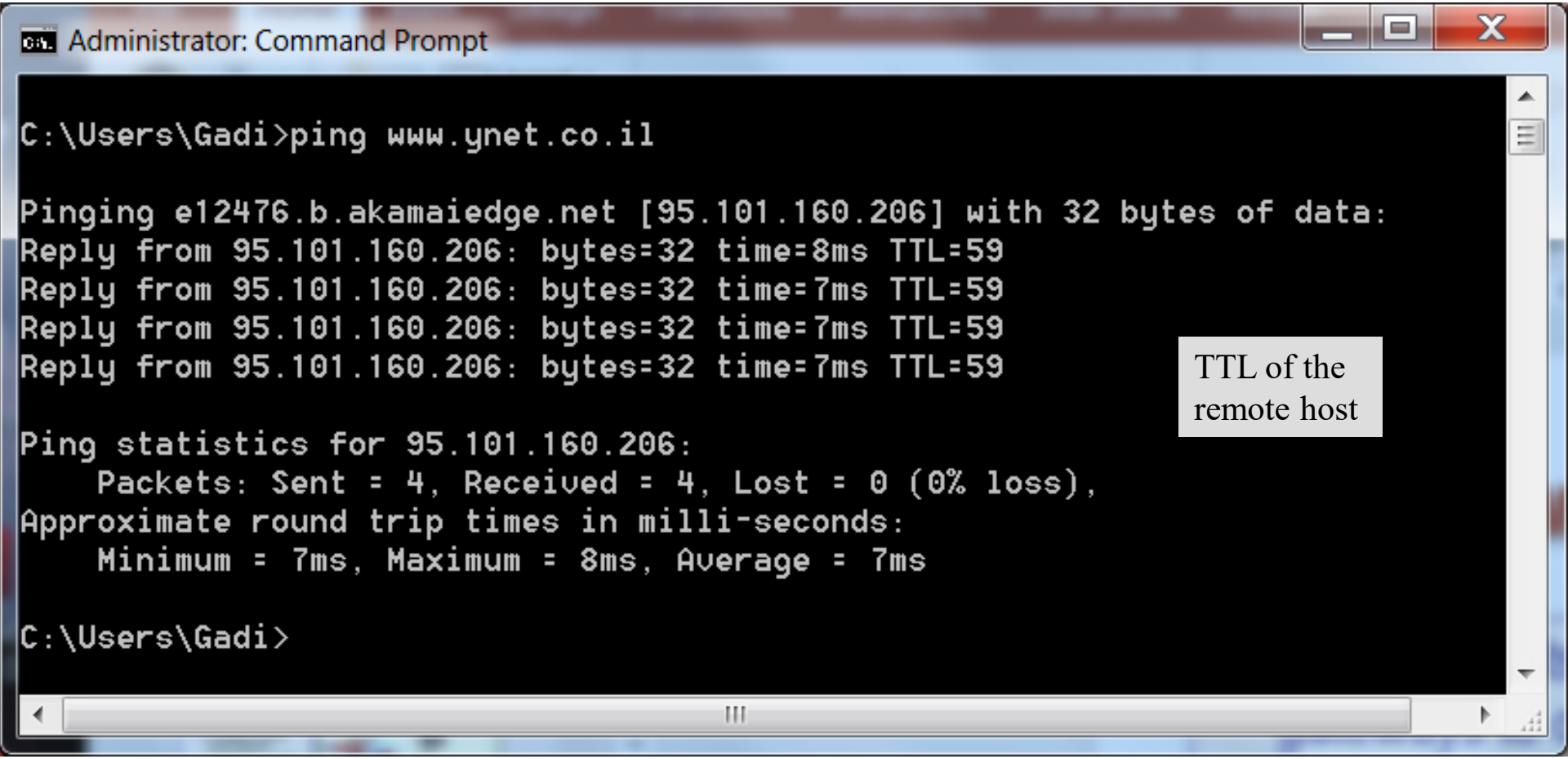
Type	Code	description
0	0	echo reply (ping)
3	0	dest. network unreachable
3	1	dest host unreachable
3	2	dest protocol unreachable
3	3	dest port unreachable
3	6	dest network unknown
3	7	dest host unknown
4	0	source quench (congestion control - not used)
8	0	echo request (ping)
9	0	route advertisement
10	0	router discovery
11	0	TTL expired
12	0	bad IP header

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PING and ICMP

Type	Code	description
8	0	echo request (ping)
0	0	echo reply (ping)



Windows XP/7/10/11 TTL=128 ; Unix/Linux/MAC TTL=64 .....

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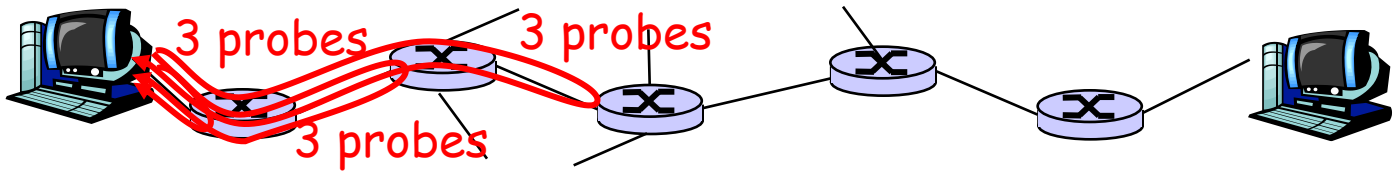
Traceroute and ICMP

- Source sends series of UDP segments to dest
  - First has TTL =1
  - Second has TTL=2, etc.
  - Unlikely port number
- When n-th datagram arrives to n-th router:
  - Router discards datagram
  - And sends to source an ICMP message (type 11, code 0) TTL expired
  - Message includes name of router& IP address

- When ICMP message arrives, source calculates RTT
- Traceroute does this 3 times

Stopping criterion

- UDP segment eventually arrives at destination host
- Destination returns ICMP "host unreachable" packet (type 3, code 3)
- When source gets this ICMP, stops.



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IP addresses: how to get one?

Hosts (host portion):

- hard-coded by system admin in a file
- DHCP: Dynamic Host Configuration Protocol: dynamically get address:
  - host broadcasts “DHCP discover” msg
  - DHCP server responds with “DHCP offer” msg
  - host requests IP address: “DHCP request” msg
  - DHCP server sends address: “DHCP ack” msg

(more shortly)

DHCP: Dynamic Host Configuration Protocol

Goal: allows host to *dynamically* obtain its IP address from network server when it joins network

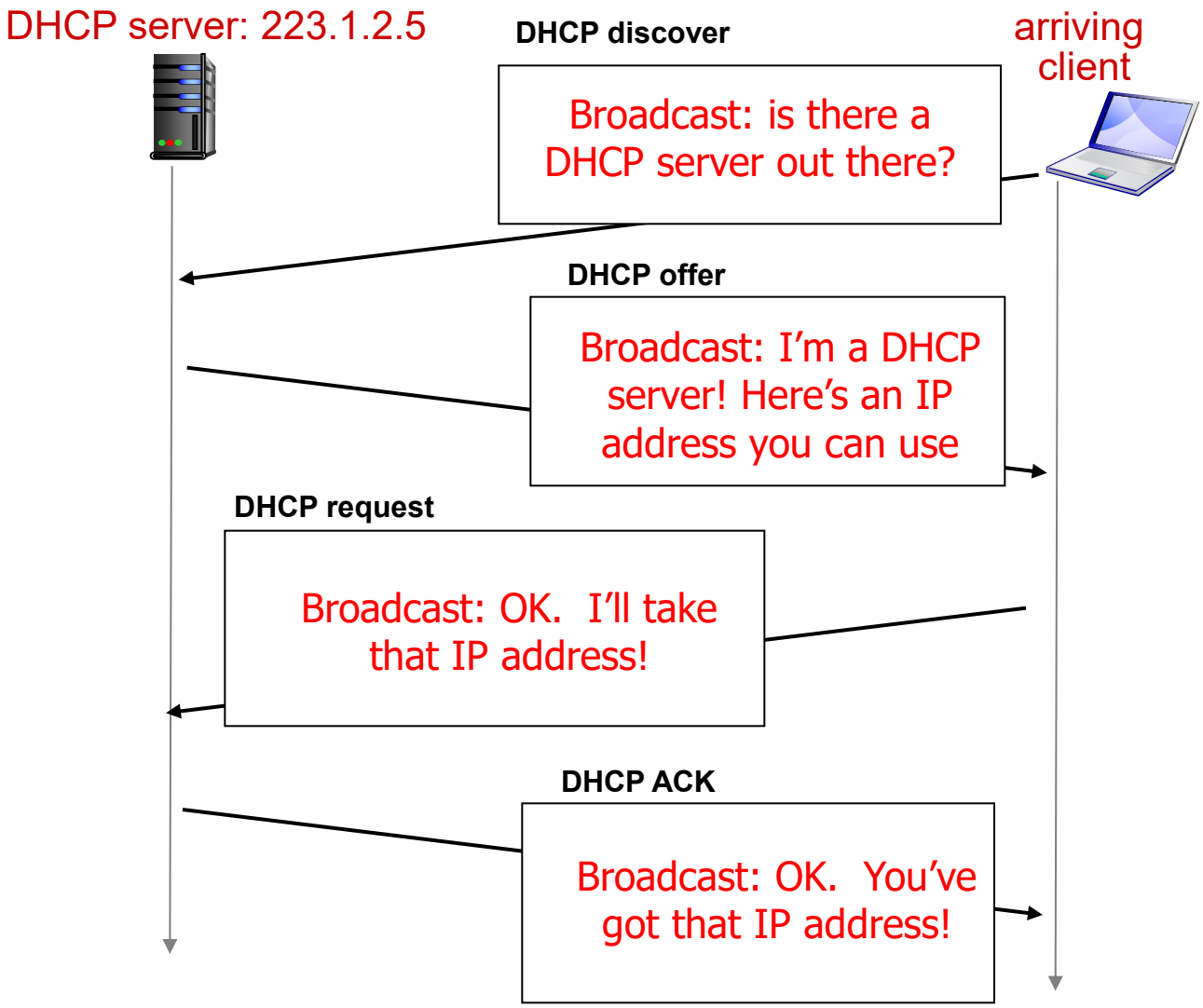
- Can renew its lease on address in use
- Allows reuse of addresses (only hold address while connected an "on"
- Support for mobile users who want to join network (more shortly)

DHCP overview

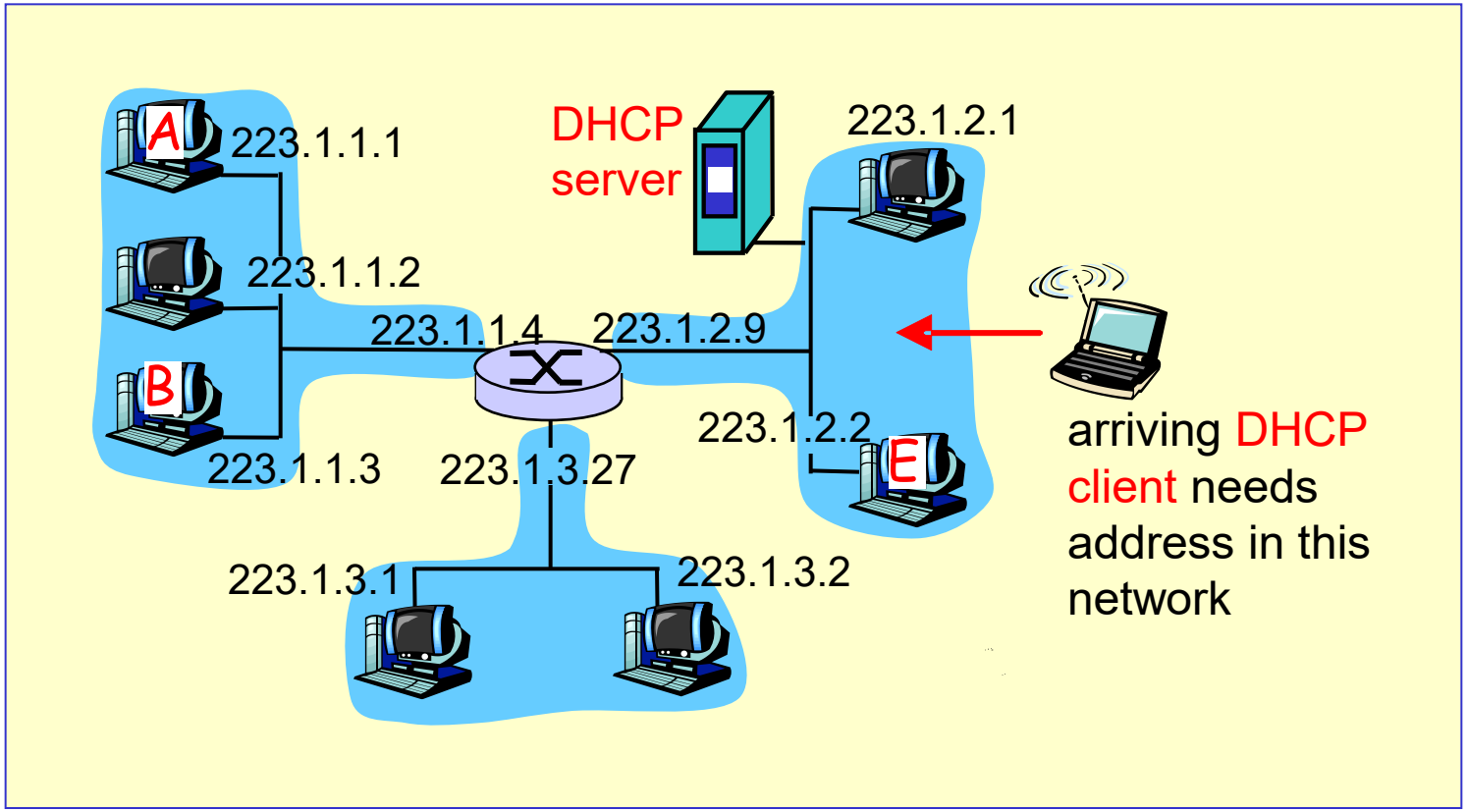
- host broadcasts "DHCP discover" msg
- DHCP server responds with "DHCP offer" msg
- host requests IP address: "DHCP request" msg
- DHCP server sends address: "DHCP ack" msg

DHCP runs over UDP (why it cannot run over TCP?)

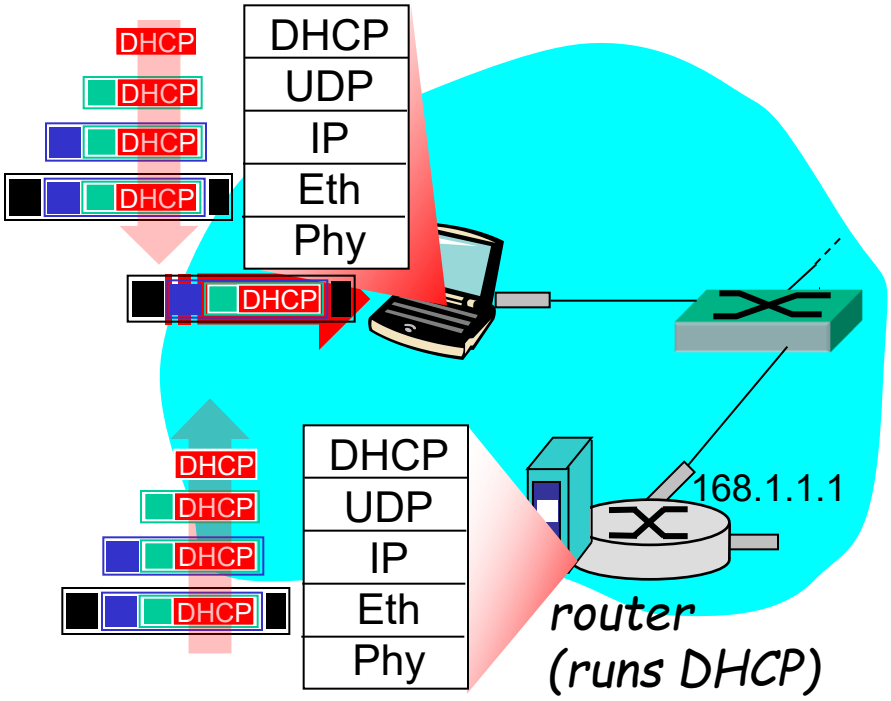
DHCP client-server scenario



DHCP client-server scenario

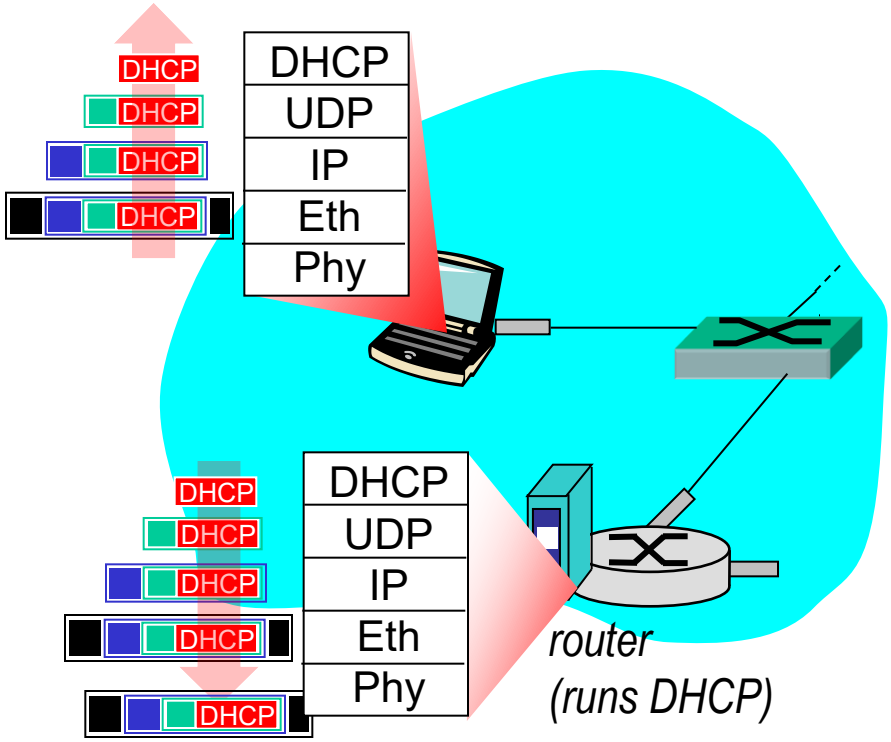


DHCP: example



- connecting laptop needs its IP address.
- DHCP request encapsulated in UDP, encapsulated in IP, encapsulated in link layer protocol (Ethernet)
- Ethernet frame broadcast (dest: FFFFFFFFFFFF) on LAN, received at router running DHCP server
- Ethernet demux'ed to IP demux'ed, UDP demux'ed to DHCP.

DHCP: example



- DHCP server formulates DHCP ACK containing client's IP address (and more information …)

- encapsulation of DHCP server, frame forwarded to client, demux'ing up to DHCP at client
- client now knows its IP address, ++

DHCP: more than IP address

DHCP returns more than just allocated IP address on subnet:

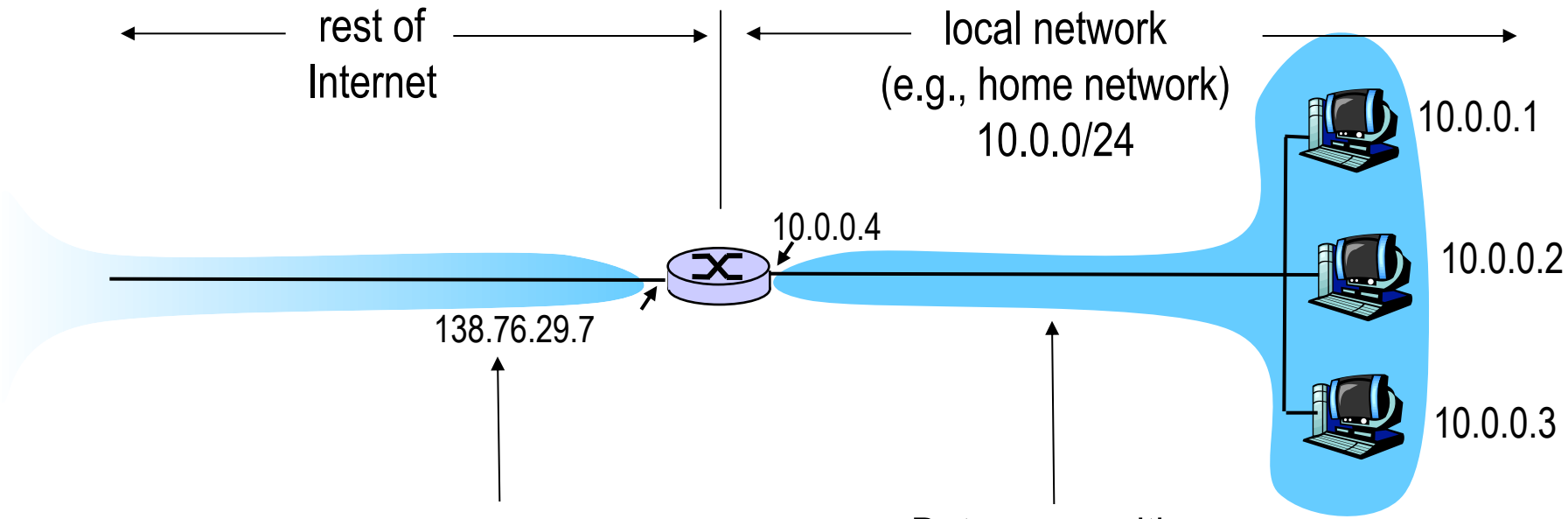
1. IP address
2. IP address of first-hop router for client
3. IP address of DNS sever
4. Network mask (indicating network versus host portion of address)



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- NAT
- IP v6

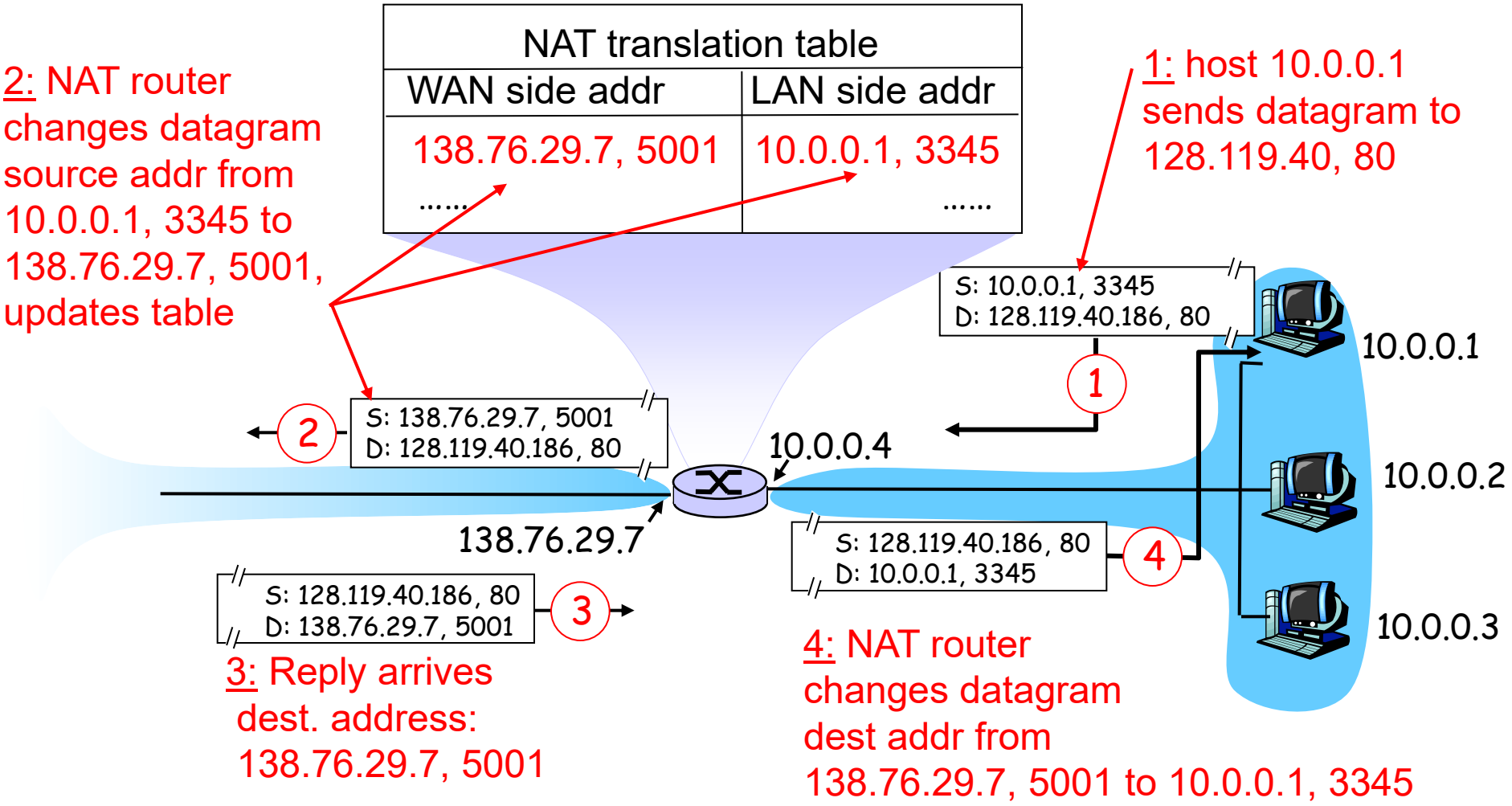
NAT: Network Address Translation



All datagrams *leaving* local network have same single source NAT IP address: 138.76.29.7, different source port numbers

Datagrams with source or destination in this network have 10.0.0/24 address for source, destination (as usual)

NAT: Network Address Translation



NAT: Network Address Translation

Motivation: local network uses just one IP address as far as outside word is concerned

- no need to be allocated range of addresses from ISP: one IP address is used for all devices
- can change addresses of devices in local network without notifying outside world
- can change ISP without changing addresses of devices in local network
- devices inside local net not explicitly addressable, visible by outside world (a security plus).

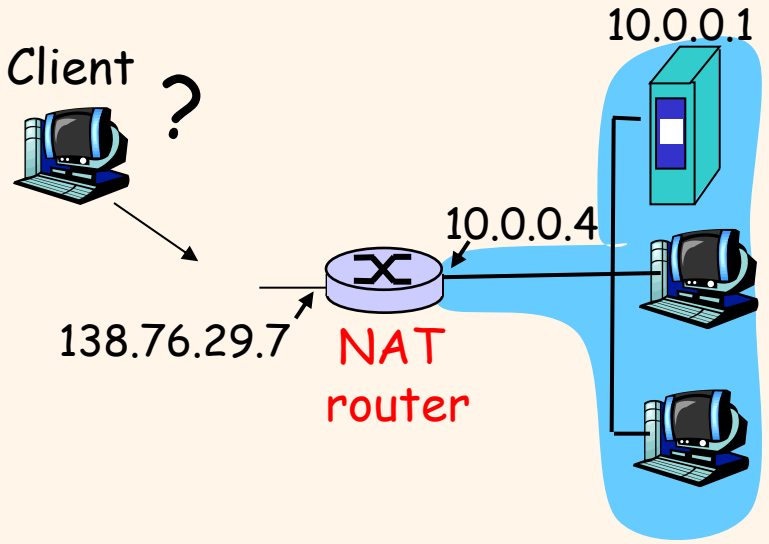
NAT: Network Address Translation

- 16-bit port-number field
  - 60,000+ simultaneous connections with a single LAN-side address!
- NAT is controversial
  - routers should only process up to layer 3
  - violates end-to-end argument
    - NAT possibility must be taken into account by app designers, eg, P2P applications
  - address shortage should instead be solved by IPv6

NAT traversal problem

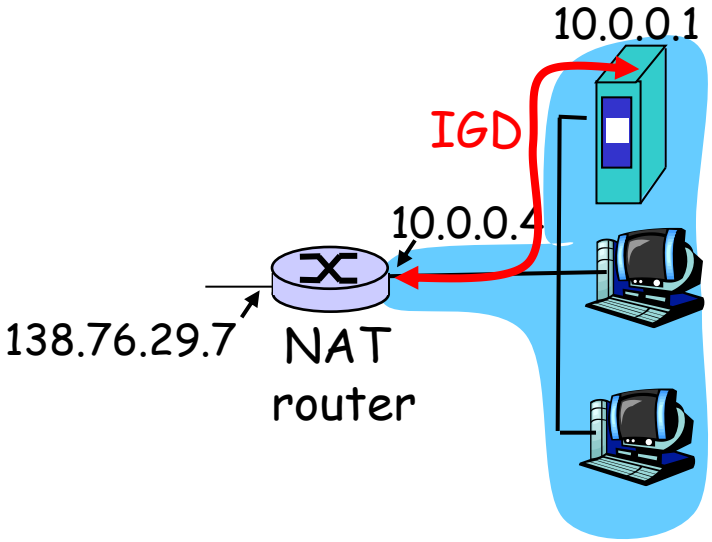
- Client want to connect to server with address 10.0.0.1
  - server address 10.0.0.1 local to LAN (client can't use it as destination addr)
  - only one externally visible NATted address: 138.76.29.7

- Solution 1: statically configure NAT to forward incoming connection requests at given port to server
  - e.g., (138.76.29.7, port 2500) always forwarded to 10.0.0.1 port 2500



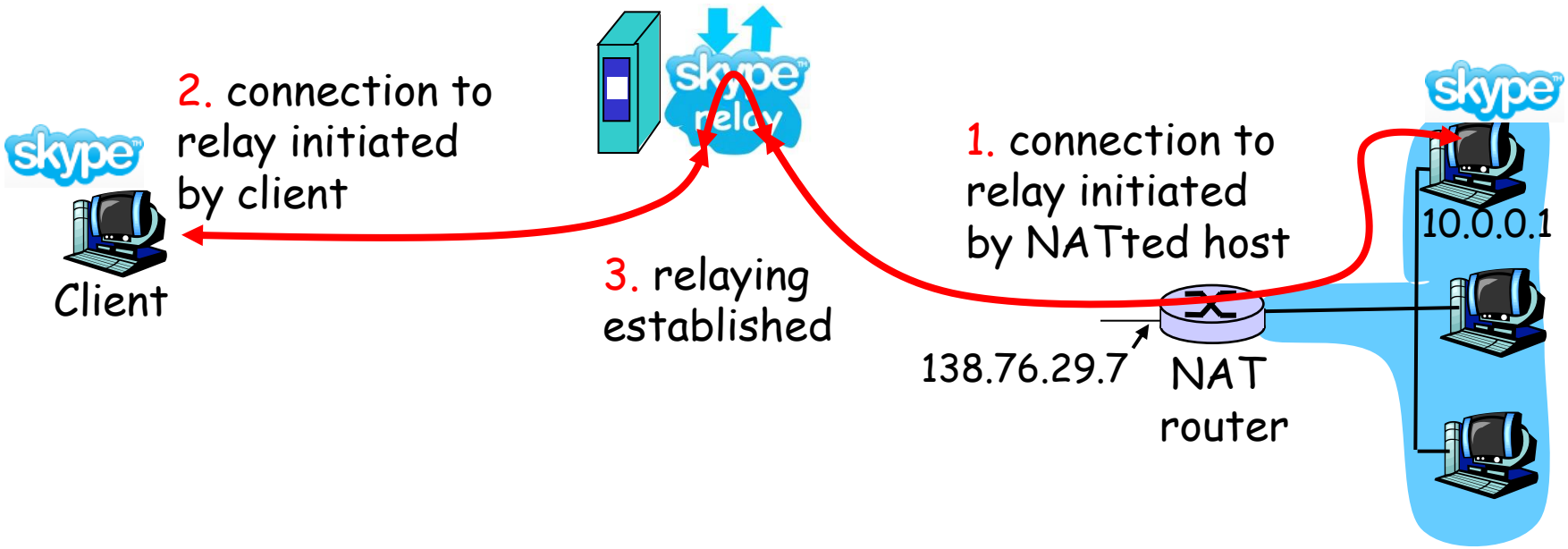
NAT traversal problem

- Solution 2: Universal Plug and Play (UPnP) Internet Gateway Device (IGD) Protocol. Allows NATted host to:
  - learn public IP address (138.76.29.7)
  - enumerate existing port mappings
  - add/remove port mappings (with lease times)



NAT traversal problem

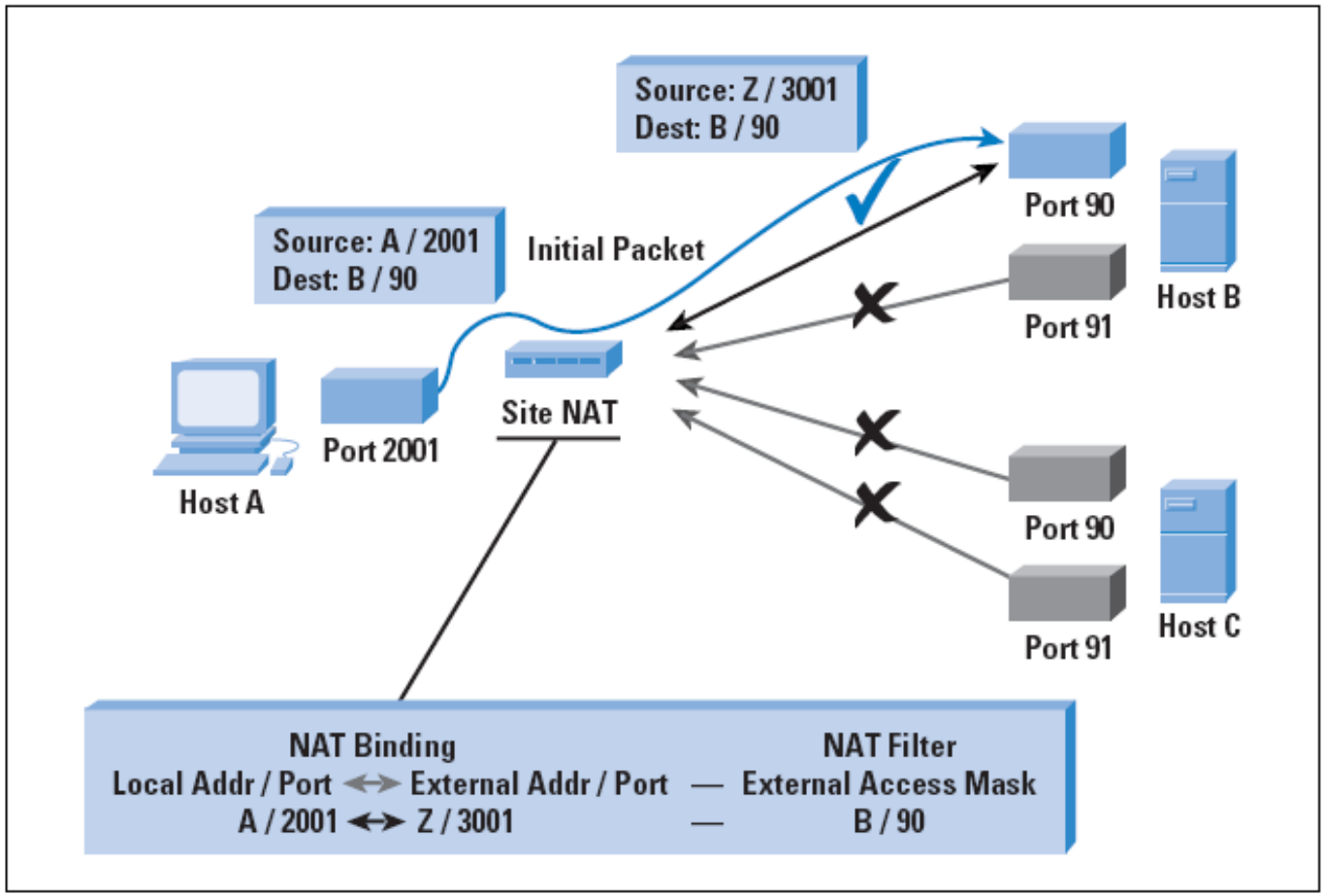
- Solution 3: relaying (used in Skype)
  - NATed server establishes connection to relay
  - External client connects to relay
  - relay bridges packets between to connections



NAT types

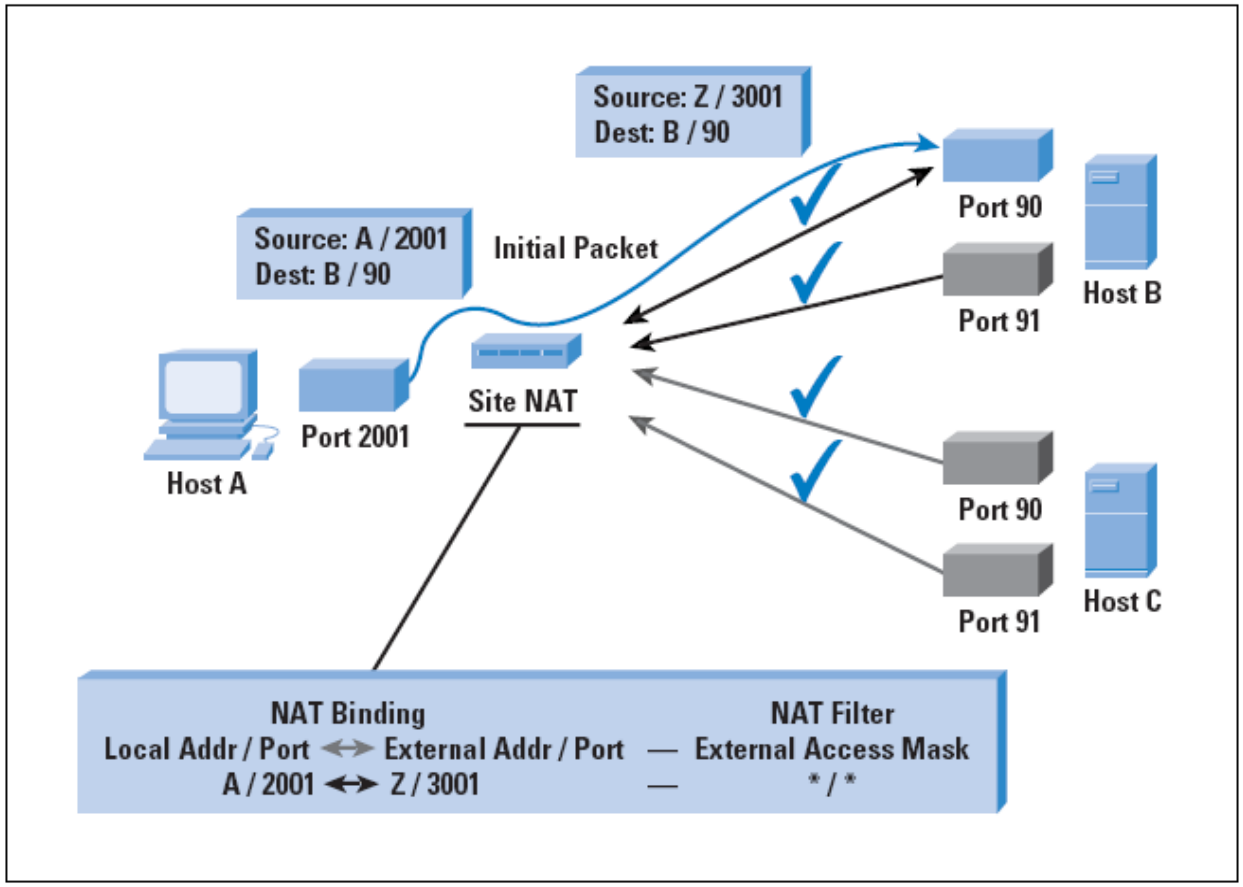
- Symmetric
- Port restricted cone
- Address restricted cone
- Full cone

Symmetric NAT
(most restrictive; rare)

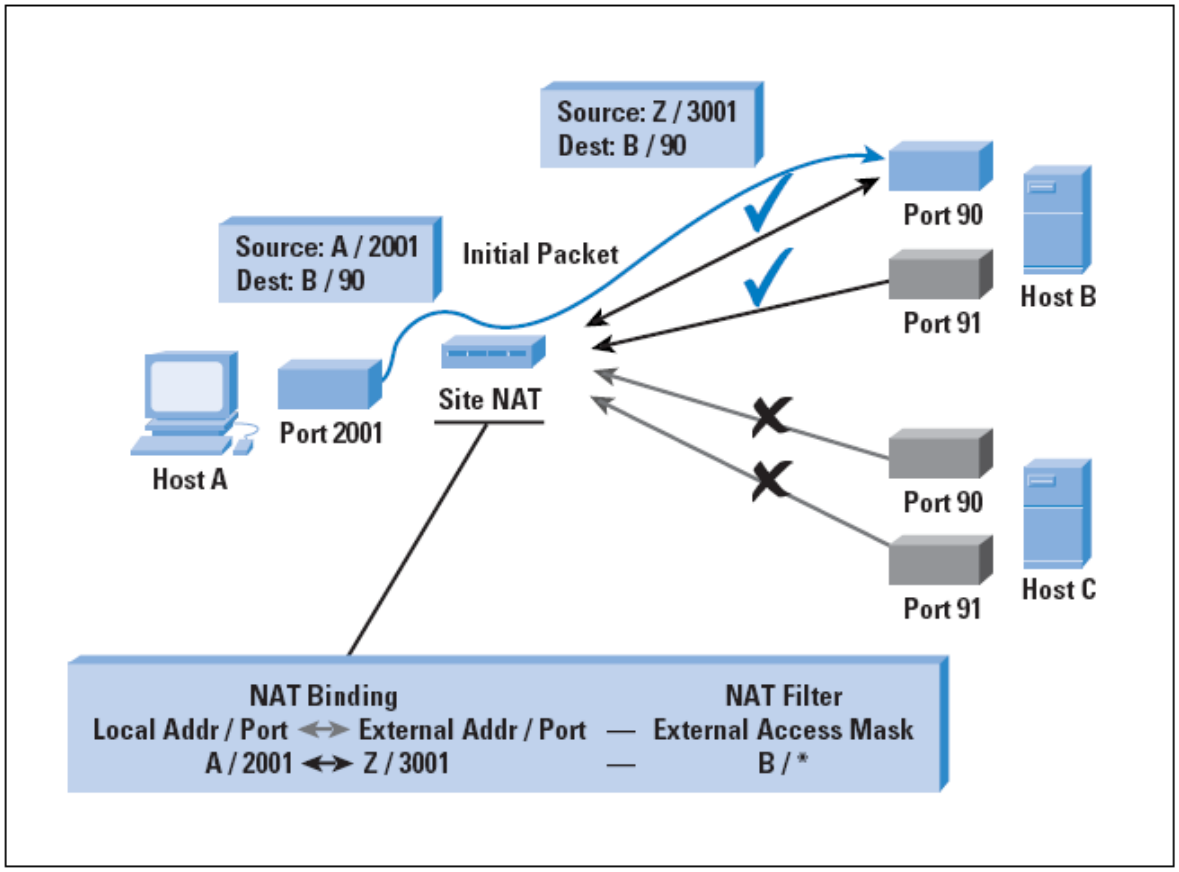


Full-cone NAT

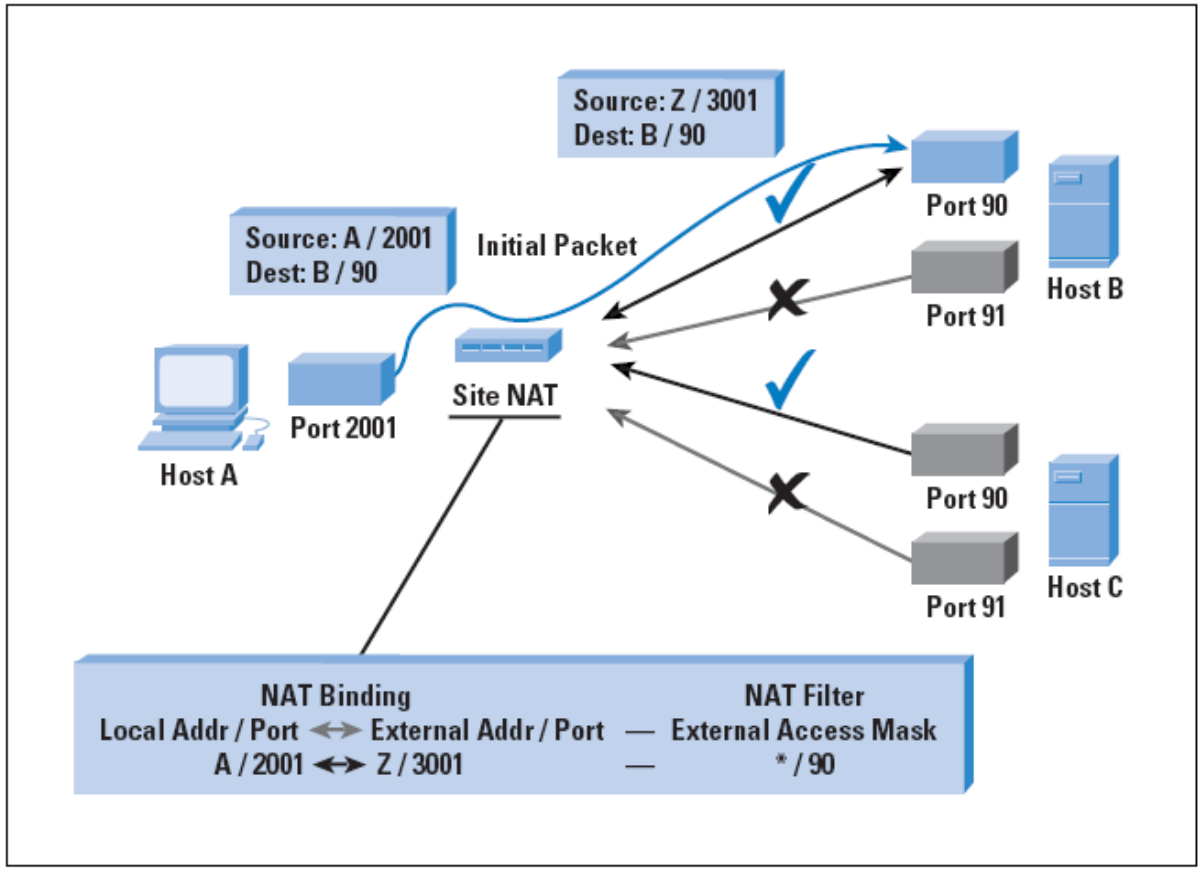
(least restrictive)



Address-restricted-cone NAT

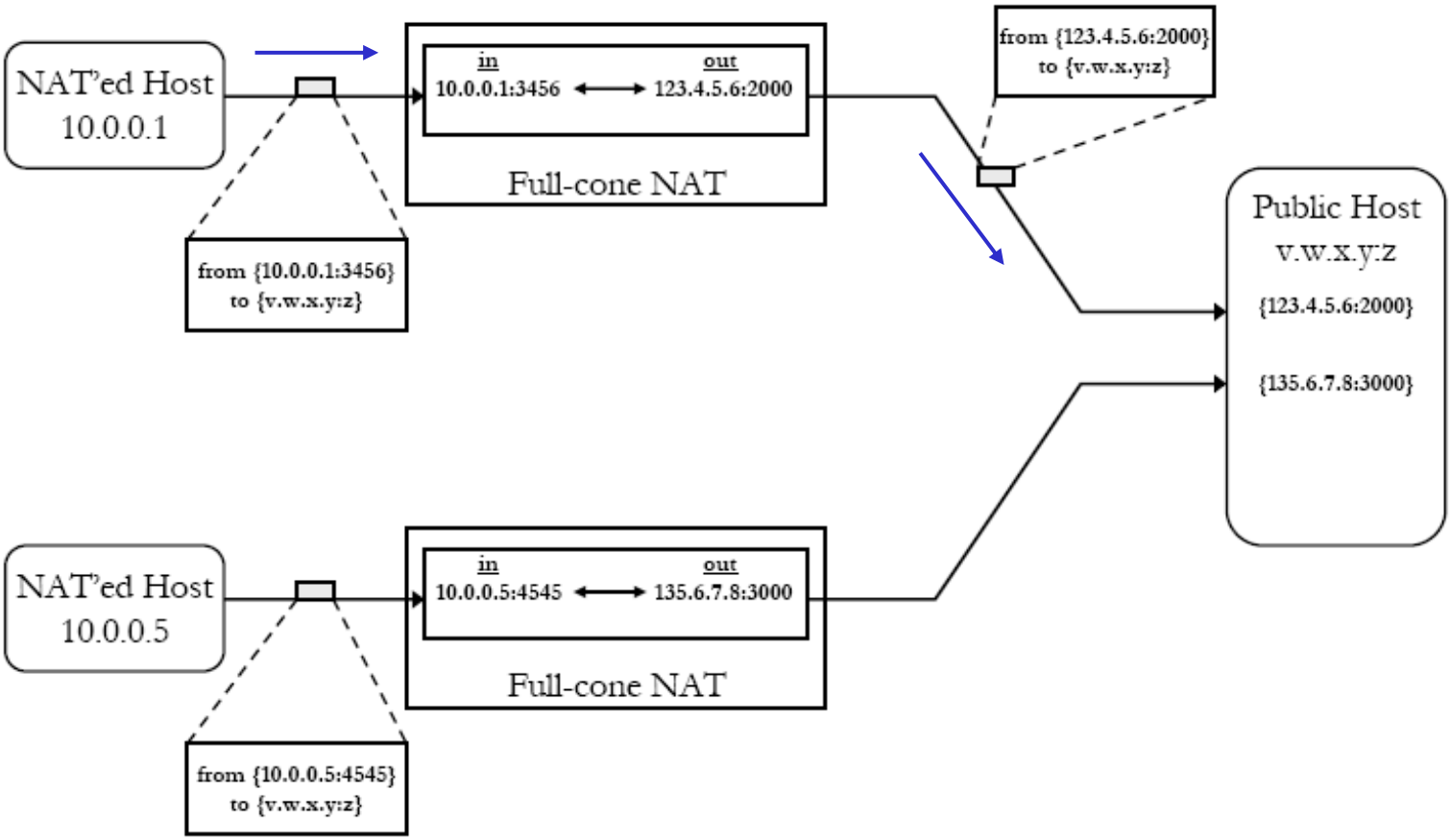


Port-restricted-cone NAT



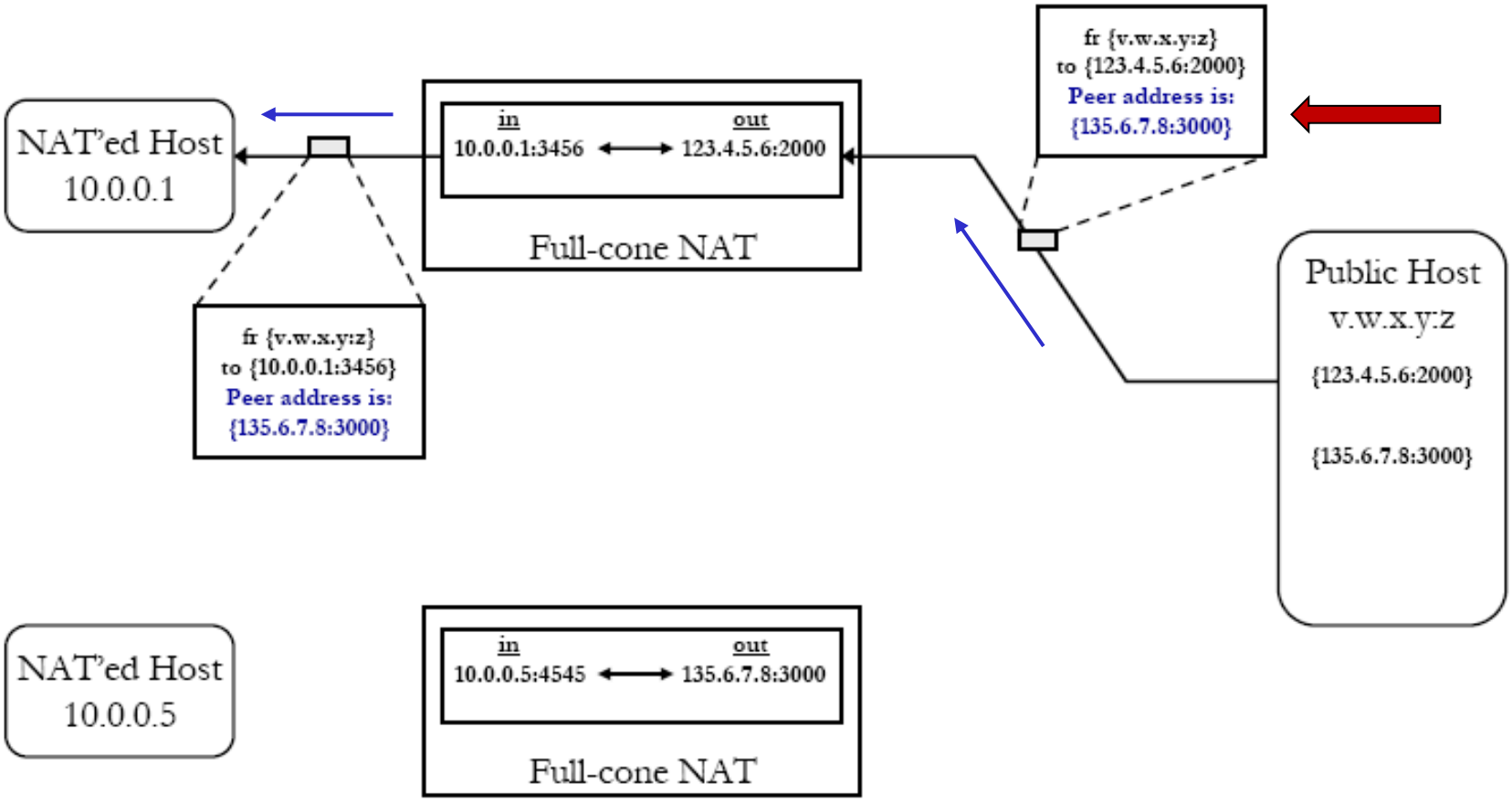
Full-cone NAT Traversal (over UDP)

Step 1 – Port Mapping



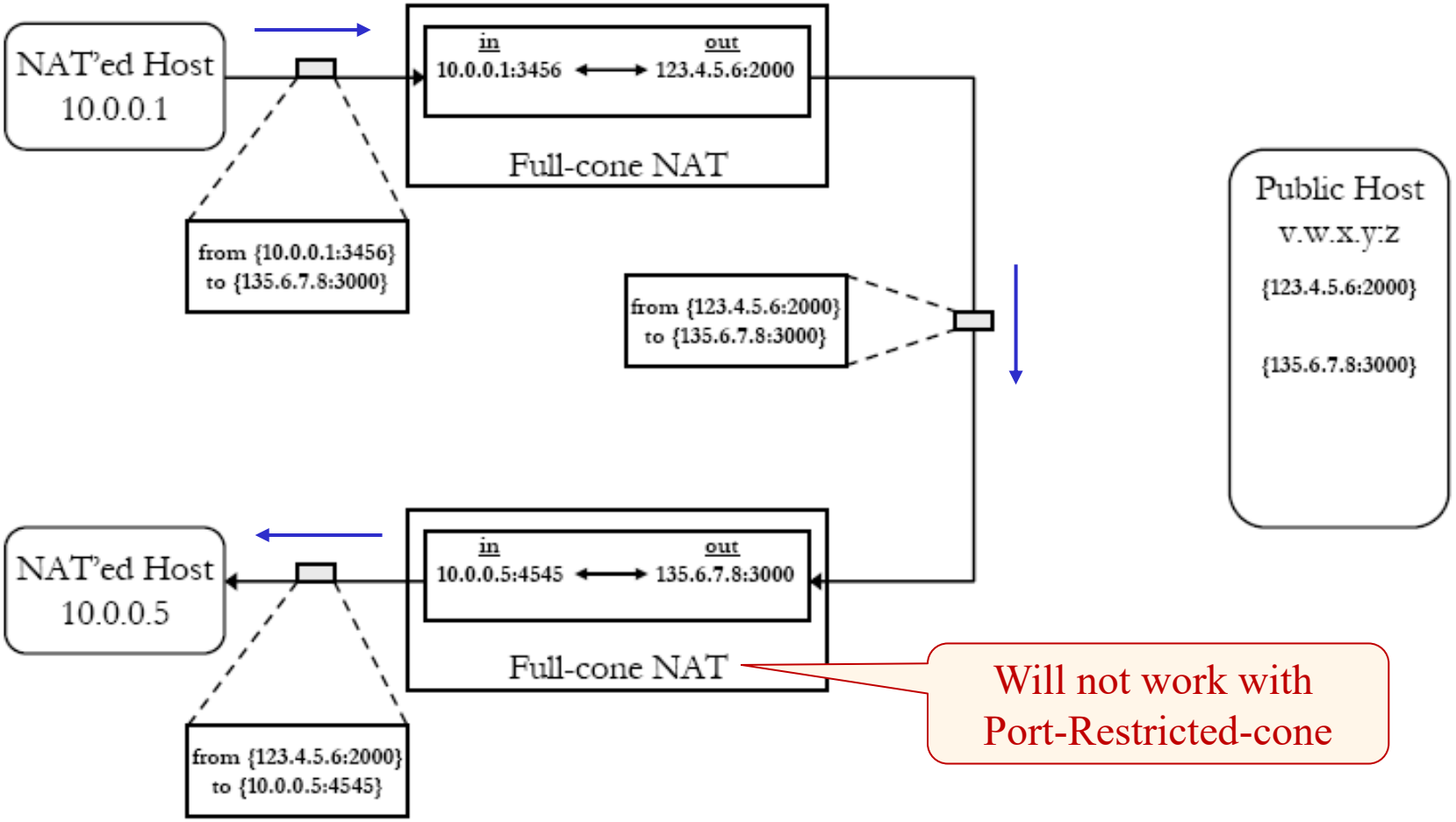
Full-cone NAT Traversal (over UDP)

Step 2 – Address Discovery



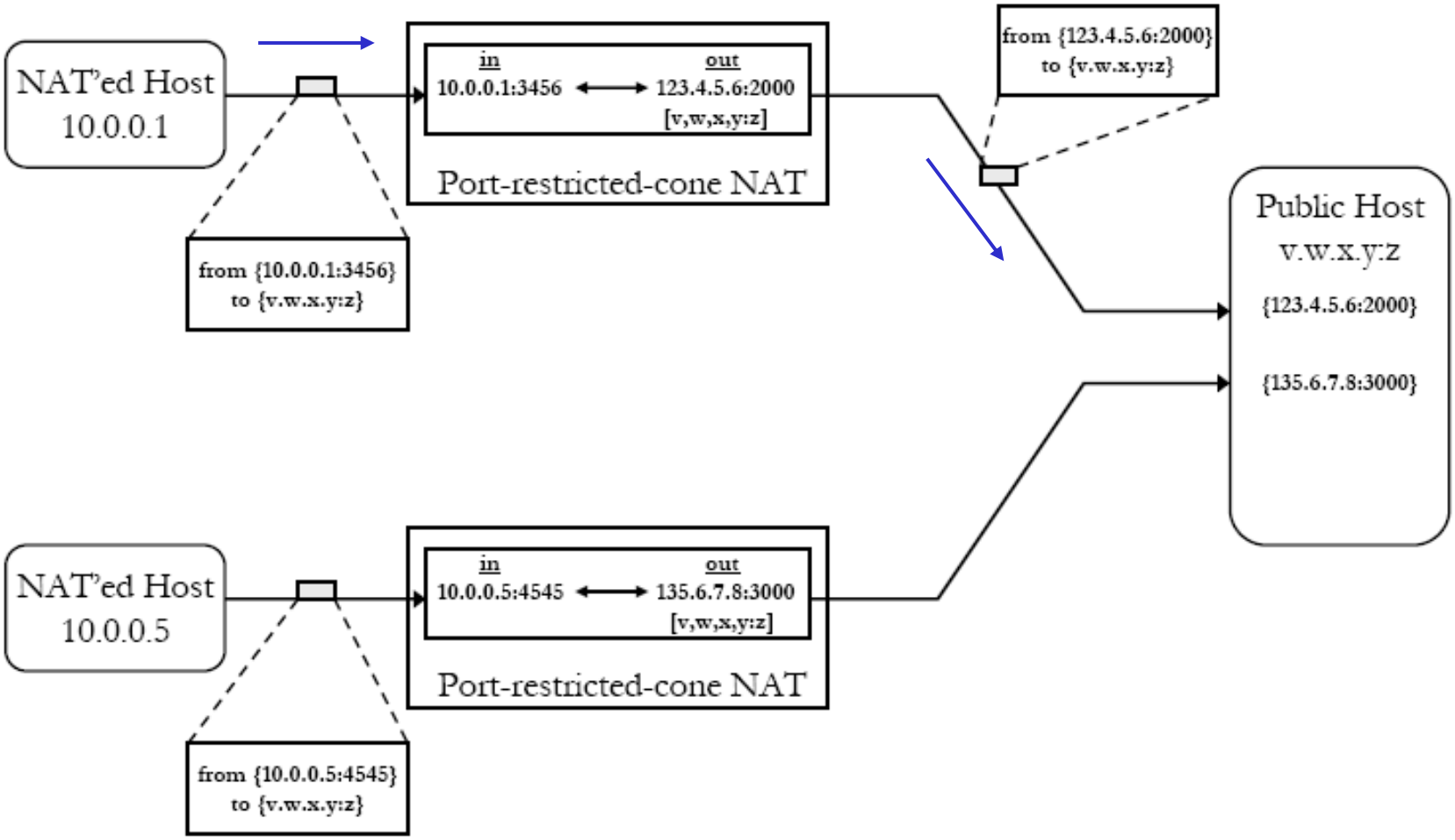
Full-cone NAT Traversal (over UDP)

Step 3 – Data Exchange



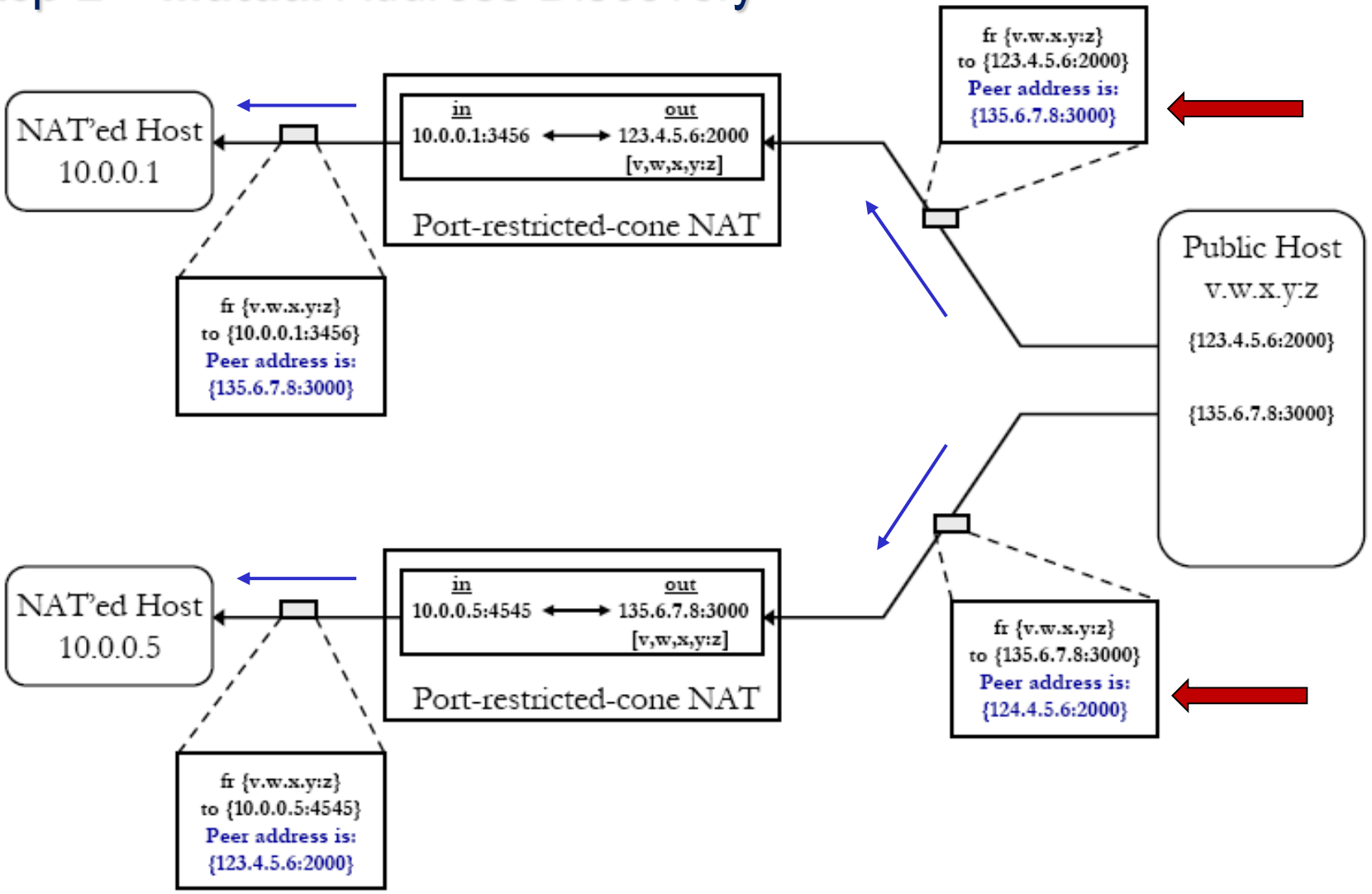
Port-restricted-cone NAT Traversal (over UDP)

Step 1 – Port Mapping



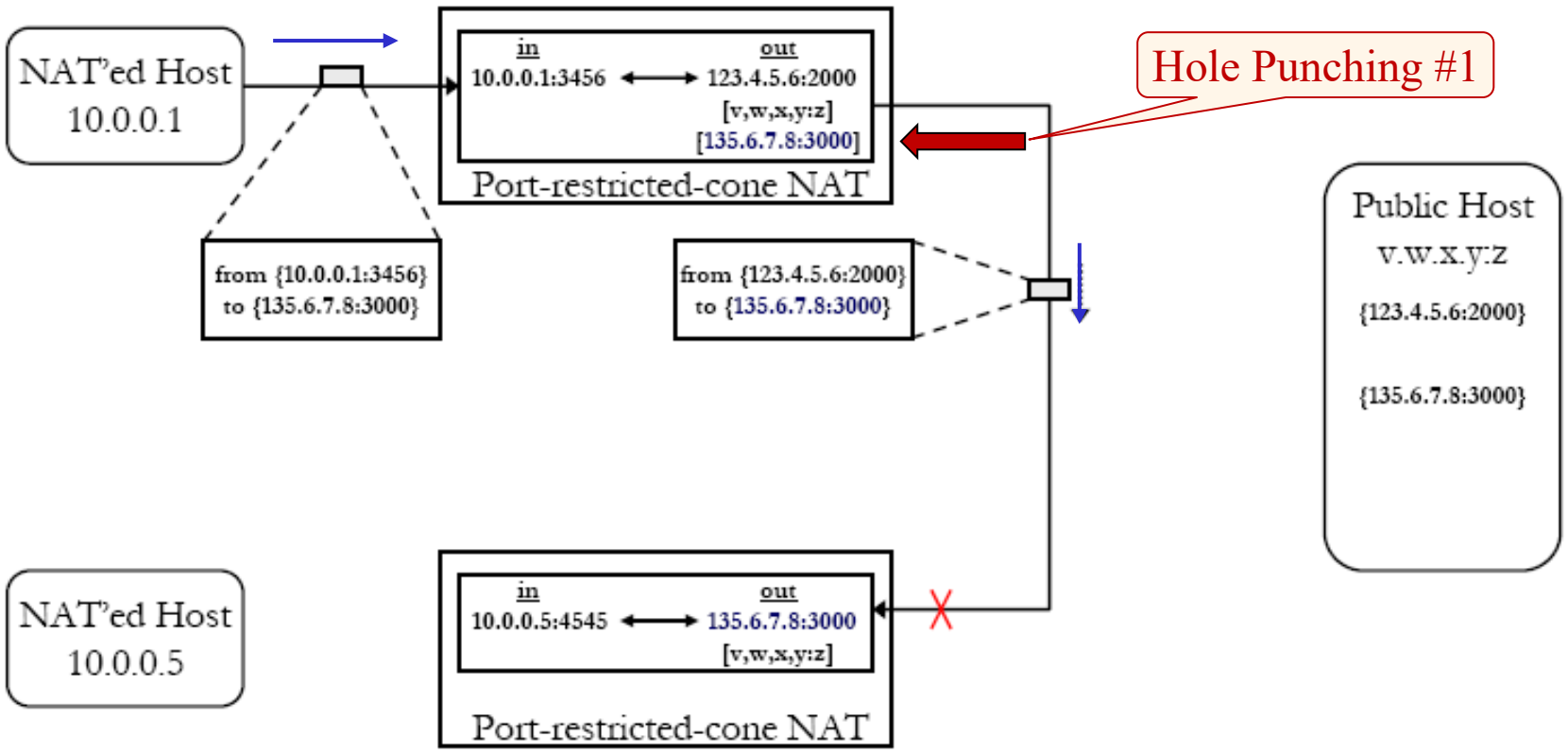
Port-restricted-cone NAT Traversal (over UDP)

Step 2 – **Mutual** Address Discovery



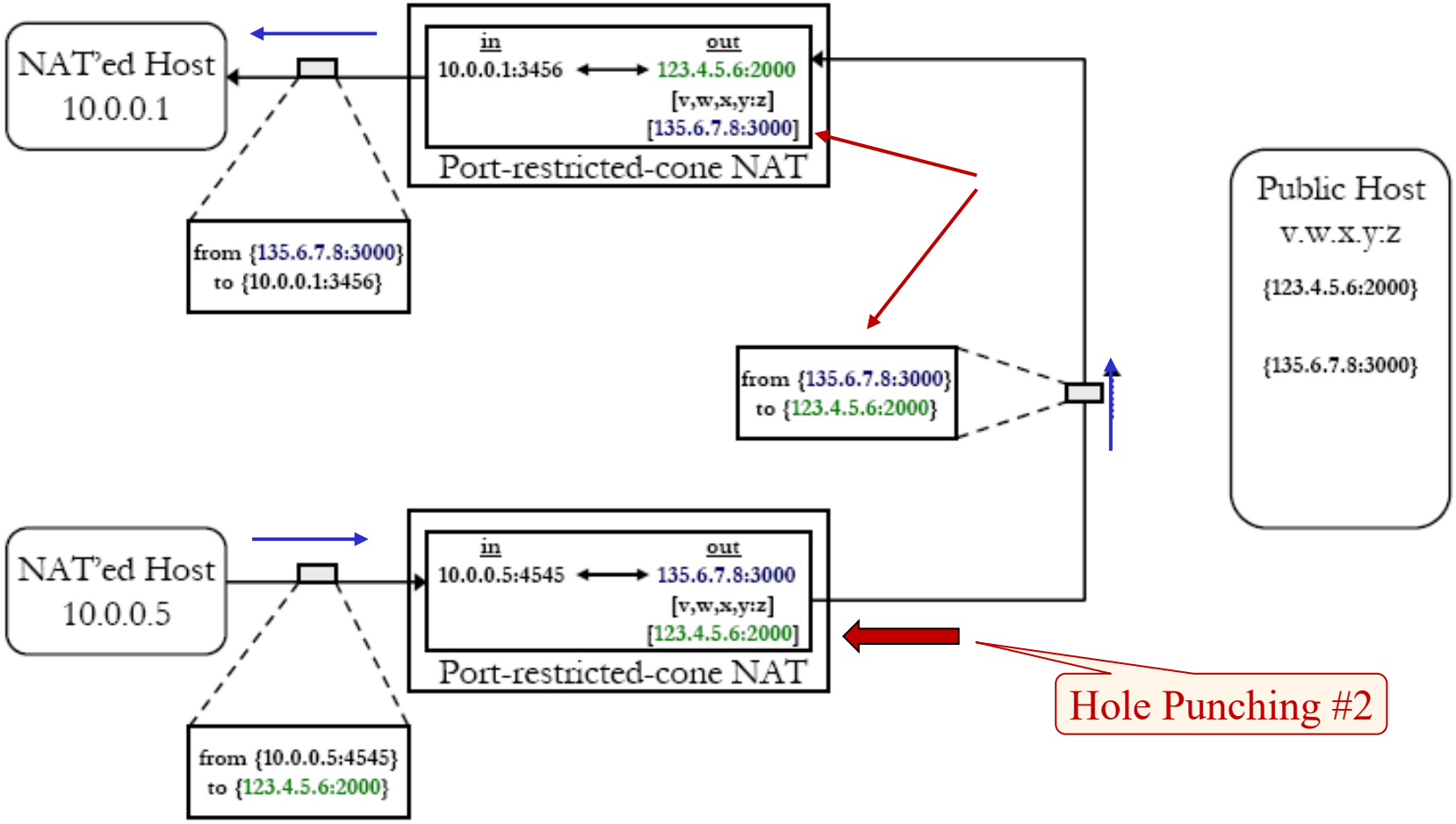
Port-restricted-cone NAT Traversal (over UDP)

Step 3 – Hole Punching #1



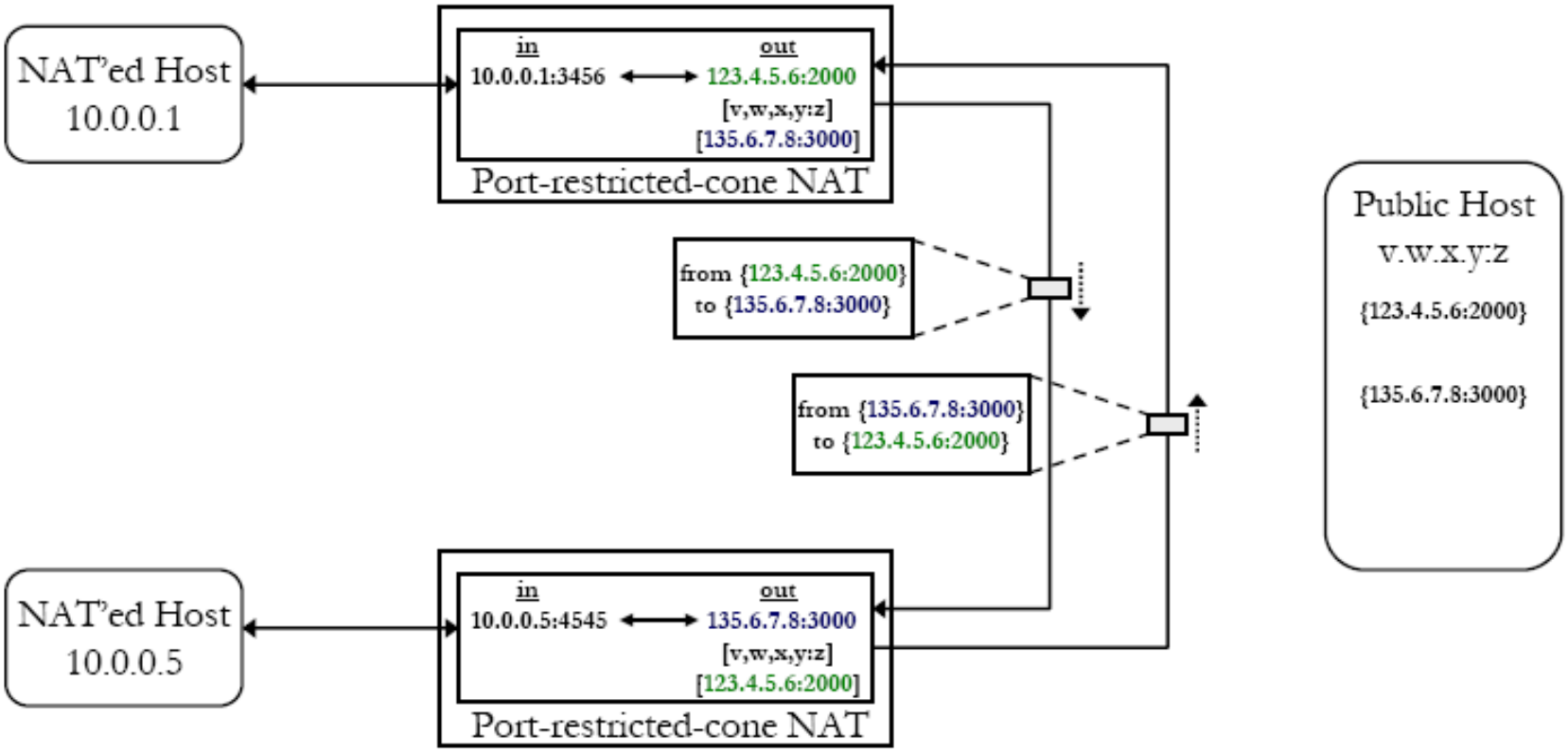
Port-restricted-cone NAT Traversal (over UDP)

Step 3 – Hole Punching #2



Port-restricted-cone NAT Traversal (over UDP)

Step 3 – Data Exchange



Today

- ✓ IP addressing introduction
- ✓ IP addressing CIDR
- ✓ How routing works over IP
- ✓ IP in details
- ✓ ICMP
  - ✓ Ping
  - ✓ Traceroute
- ✓ DHCP
- ✓ NAT
- IP v6

Advanced Topic: IPv6

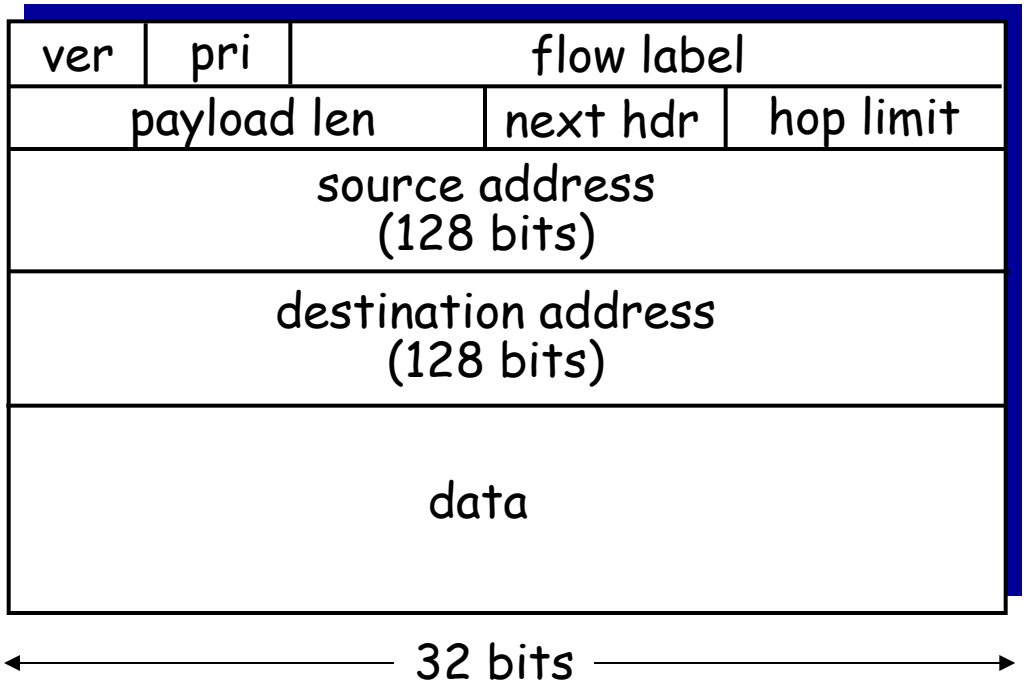
- Initial motivation
  - 32-bit address space completely allocated by 2008, 2018,...
- Additional motivation
  - header format helps speed processing/forwarding
  - header changes to facilitate QoS
  - new “anycast” address: route to “best” of several replicated servers
- IPv6 datagram format
  - fixed-length 40 byte header

IPv6 Header (Cont)

Priority: identify priority among datagrams in flow

Flow Label: identify datagrams in same "flow."
(concept of"flow" not well defined).

Next header: identify upper layer protocol for data

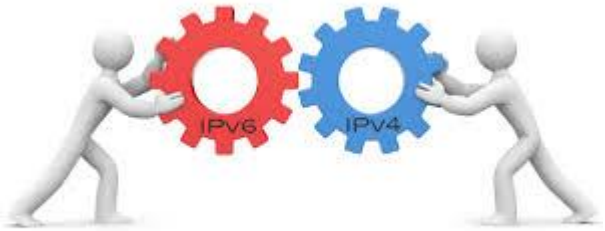


Other Changes from IPv4

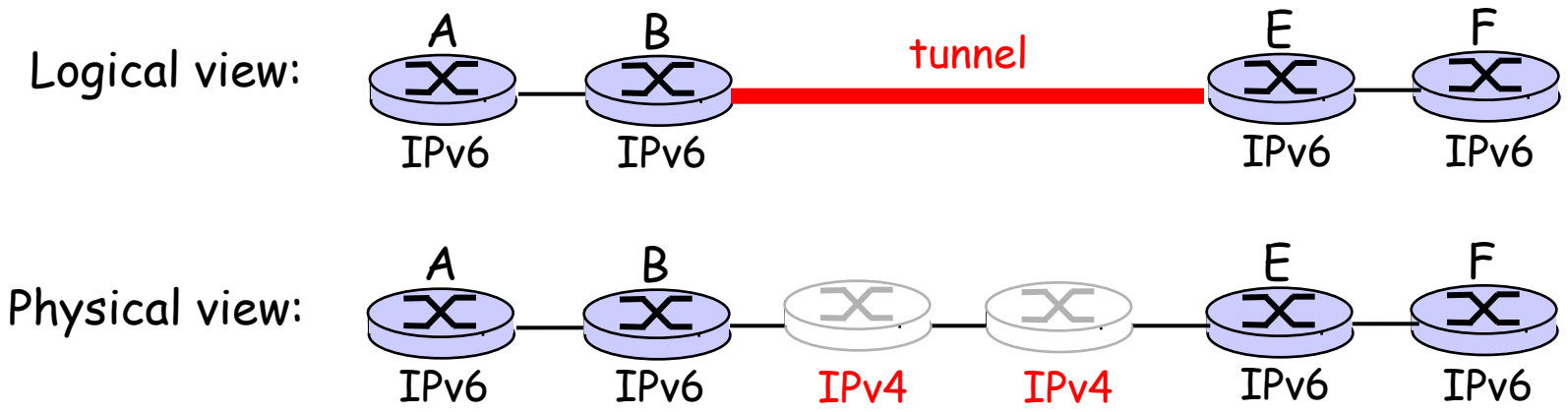
- *Checksum*: removed entirely to reduce processing time at each hop
- No fragmentation allowed: large IPv6 datagram are not divided ("fragmented") within net
- *ICMPv6*: new version of ICMP
  - additional message types, e.g. "Packet Too Big"
  - multicast group management functions

Transition From IPv4 To IPv6

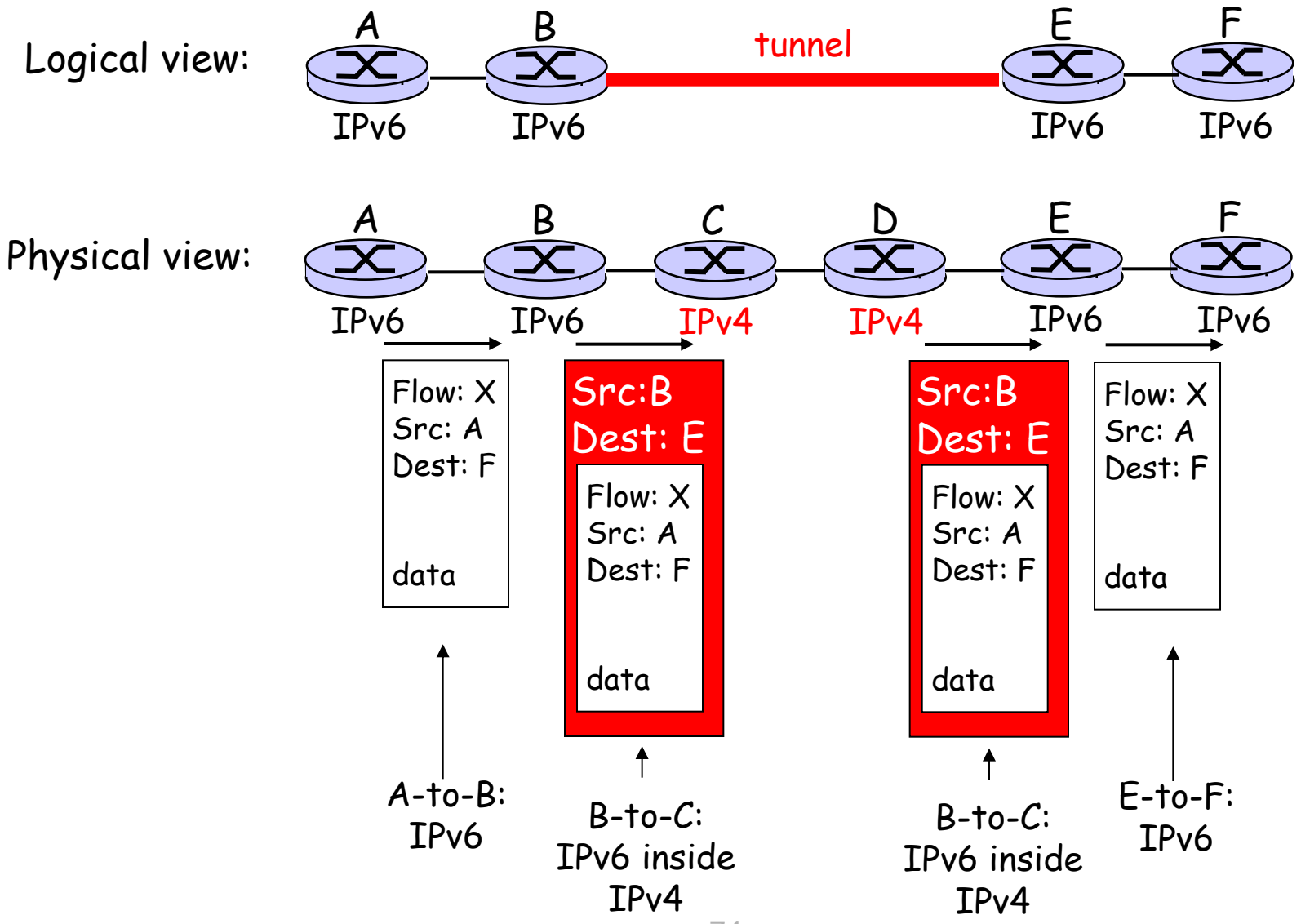
- Not all routers can be upgraded simultaneous
  - no “flag days”
  - How will the network operate with mixed IPv4 and IPv6 routers?
- *Tunneling:* IPv6 carried as payload in IPv4 datagram among IPv4 routers



Tunneling


Logical view:
A
B
tunnel
E
F
IPv6
IPv6
IPv6
IPv6
Physical view:
A
B
E
F
IPv6
IPv6
IPv4
IPv4
IPv6
IPv6

Tunneling

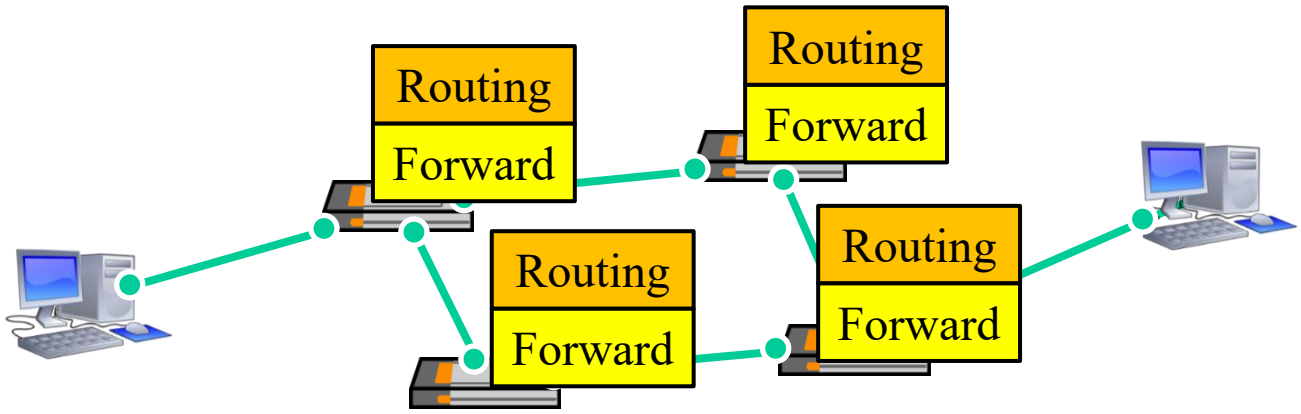

Logical view:
A
B
tunnel
E
F
IPv6
IPv6
IPv6
IPv6
Physical view:
A
B
C
D
E
F
IPv6
IPv6
IPv4
IPv4
IPv6
IPv6
Flow: X
Src: A
Dest: F
data
Src:B
Dest: E
Flow: X
Src: A
Dest: F
data
Src:B
Dest: E
Flow: X
Src: A
Dest: F
data
Flow: X
Src: A
Dest: F
data
A-to-B:
IPv6
B-to-C:
IPv6 inside
IPv4
B-to-C:
IPv6 inside
IPv4
E-to-F:
IPv6

IPv6: adoption

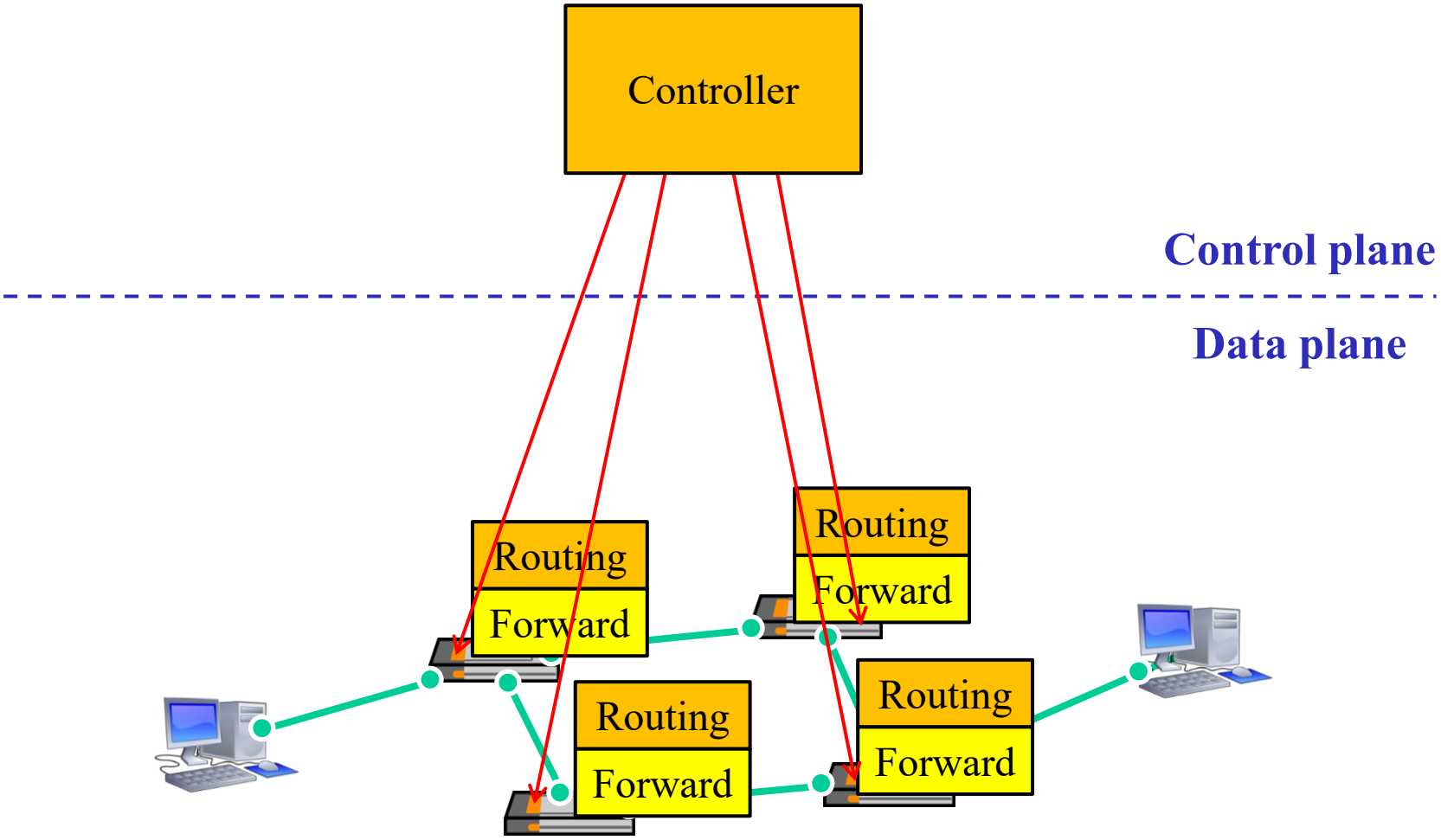
- IPv6 Capable Rate by country (%)
  - RU: 14%, Israel: 62%; US 55% India 78% (Dec. 2024) https://stats.labs.apnic.net/ipv6
- Google
  - 42% of clients access services via IPv6 (Dec. 2024) https://www.google.com/intl/en/ipv6/statistics.html
- *Long (long!) time for deployment,*
  - 20 years and counting!
  - think of application-level changes in last 20 years: Facebook, …



Software Defined Network



Software Defined Network



Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

The IP checksum applies to the IP headers and payload.

- FALSE
- TRUE

A question from an exam
Circle the correct answer

שאלה מבחינה
הקף בעיגול את התשובה המתאימה

What is the correct relation?

1. ICMP runs over UDP.
2. BGP runs over UDP.
3. RIP runs over TCP.
4. OSPF runs over TCP.
5. All of the above.
6. None of the above.

מה היחס הנכון?

1. ICMP רץ מעל UDP.
2. BGP רץ מעל UDP.
3. RIP רץ מעל TCP.
4. OSPF רץ מעל TCP.
5. כל התשובות (למעלה) נכונות.
6. כל התשובות (למעלה) לא נכונות.

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

חבילה המפוצלת על ידי נתב מורכבת מחדש אך ורק בכתובת היעד.

1. נכון
2. לא נכון

A packet fragmented by a router is re-assembled only at the destination address.

- True
- False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

שימוש ב- NAT מאפשר ללא יותר מ- 1024 אפליקציות ברשת פנימית להתקשר עם הרשת החיצונית על ידי שימוש בכתובת IP חיצונית אחת.

1. נכון
2. לא נכון

The use pf NAT allows no more than 1024 applications in the internal network to communicate with the exterior network by using just one exterior IP address.

- True
- False

Network Layer: summary

Topics:

- Network layer services
- Routing principles: link state and distance vector
- Hierarchical routing
- Internet routing protocols RIP, OSPF, IGRP, BGP
- IPv4
- ICMP, DHCP, NAT
- IPv6
- SDN

Today

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